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B.Sc. Ag. (Hons.)
Natural Farming

Complete Syllabus (Semester I - VIII)

Institute of Agriculture and Natural Sciences (IANS)
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Shared by: Aditya Kumar

M.Sc. (Ag.) Soil Science - Gold Medalist, DDUGU Gorakhpur

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Syllabus for Undergraduate Programme B.Sc. Ag. (Hons.) – Natural Farming



Institute of Agriculture and Natural Sciences (IANS)

Deen Dayal Upadhyay Gorakhpur University

Civil Lines, Gorakhpur (U.P.)-273009, INDIA

Website: https://ddugu.ac.in

Submission Certificate

A committee under the chairmanship of Dr. Anupam Mishra, Vice Chancellor, Central Agricultural University, Imphal was constituted vide F.No. 5/2/2022-EQR by Indian Council of Agricultural Research to develop syllabus for new Undergraduate Degree Programme on Natural Farming. The draft syllabus was recommended by the committee with due deliberations from the relevant stakeholders. Draft syllabus was approved by Hon'ble Minister of Agriculture & Farmers Welfare, Government of India.

Consequently, a sub-committee was constituted to realign and restructure the draft syllabus as per framework of National Education Policy-2020 and Sixth Deans' Committee. Sub-committee through 5 virtual meetings followed by physical meeting during 26-27 September, 2023 finalized the course schedule for different semesters along with credit load and content for each course with objectives, specific objectives, teaching methods and learning outcome. Detailed theory and practical schedule is also given.

The necessary provisions have been incorporated in the final syllabus for foundation courses, common courses, skill enhancement courses, elective and online choice based courses by the sub- committee constituted for the purpose. The entry, exit options (Certificate, Diploma and Degree) as per the norms of NEP-2020 and framework of Sixth Deans' Committee are now included in the final syllabus.

Undersigned Members and Chairman are pleased to submit the realigned and restructured syllabus for Undergraduate Degree in B.Sc. (Hons.) Natural Farming.

(N. Ravisankar)
Member

(Mahesh Chander)
Member

(Latika Sharma)
Member

(Sunil Pareek)
Member

(M. Suganthy)
Member

(U. K. Behera)
Member

(Dwipendra Thakuria)
Member

(E. Somasundaram)
Member

Mahendra Singh Sevda
Member

Vanita Jain
Member Secretary

Academic Programme

4 Years B.Sc. Ag. (Hons.) – Natural Farming

IV. Eligibility Criteria:

- The minimum eligibility requirement for admission in undergraduate programme on Natural Farming will be the same as in case of B.Sc. Ag. (Hons.) and allied subject.
- The draft syllabus was recommended by the committee with due deliberations from the relevant stakeholders as per National Education Policy-2020 and Sixth Deans Committee. Emphasis has been given on Skill Enhancement Courses (SEC), Experiential Learning (EL), RAWE (Rural Agricultural Work Experience), Internship, Industry Attachment as a component of Student READY as well as flexibility in choice of courses via elective courses and also through MooC online courses along with provision of project work and research internship.

HIGHLIGHTS

- The whole course program of B.Sc. Ag (Hons.) Natural Farming is of 166+ (*NCMOOC) = 176 credit hours. The minimum eligibility requirement for admission in undergraduate program on Natural Farming will be the same as in case of B.Sc. Agriculture (Hons.). After the admission of students in the university, the students will register for the Foundation Course of 3 weeks duration in the 1st semester. The course will include discussions on operational framework of academic process in university, sessions from alumni, business leaders, University academic and research managers and classes on personality development (instilling life and social skills, social awareness, ethics and values, team work, knowing and being, leadership, etc.) and communication skills. Steps will be taken to identify the strength and weakness of students (with remedial measures), diverse potentialities will be taken into account and cultural integration of students from different backgrounds will be encouraged. It will also create a platform for students to learn from each other's life experiences.
- The first year of the course is dedicated for knowledge and skill enhancement in Natural Farming including Indian heritage and applied aspects of natural farming. After satisfactory completion of 40 credits of courses in two semesters and subsequent satisfactory completion of 10 credits (10 weeks) of industry/ institute training/ internship, the student will become eligible for the award of UG-Certificate in Natural Farming on exit. The students continuing the study further, would not have to attend the internship after 1st year.
- The skill program during the first year is termed as Skill Enhancement Courses (SEC). The students will have flexibility and choice in selection of skill development area from a basket of SEC modules to be offered by the parent institute. After one- week common orientation on different skill enhancement modules, students will take up either one or more modules (maximum 4) as per the local needs and gain complete hands-on experience on these modules. In addition to the modules proposed in this report, the SAUs can formulate other modules relevant to the respective regions or modify the titles of the proposed modules.

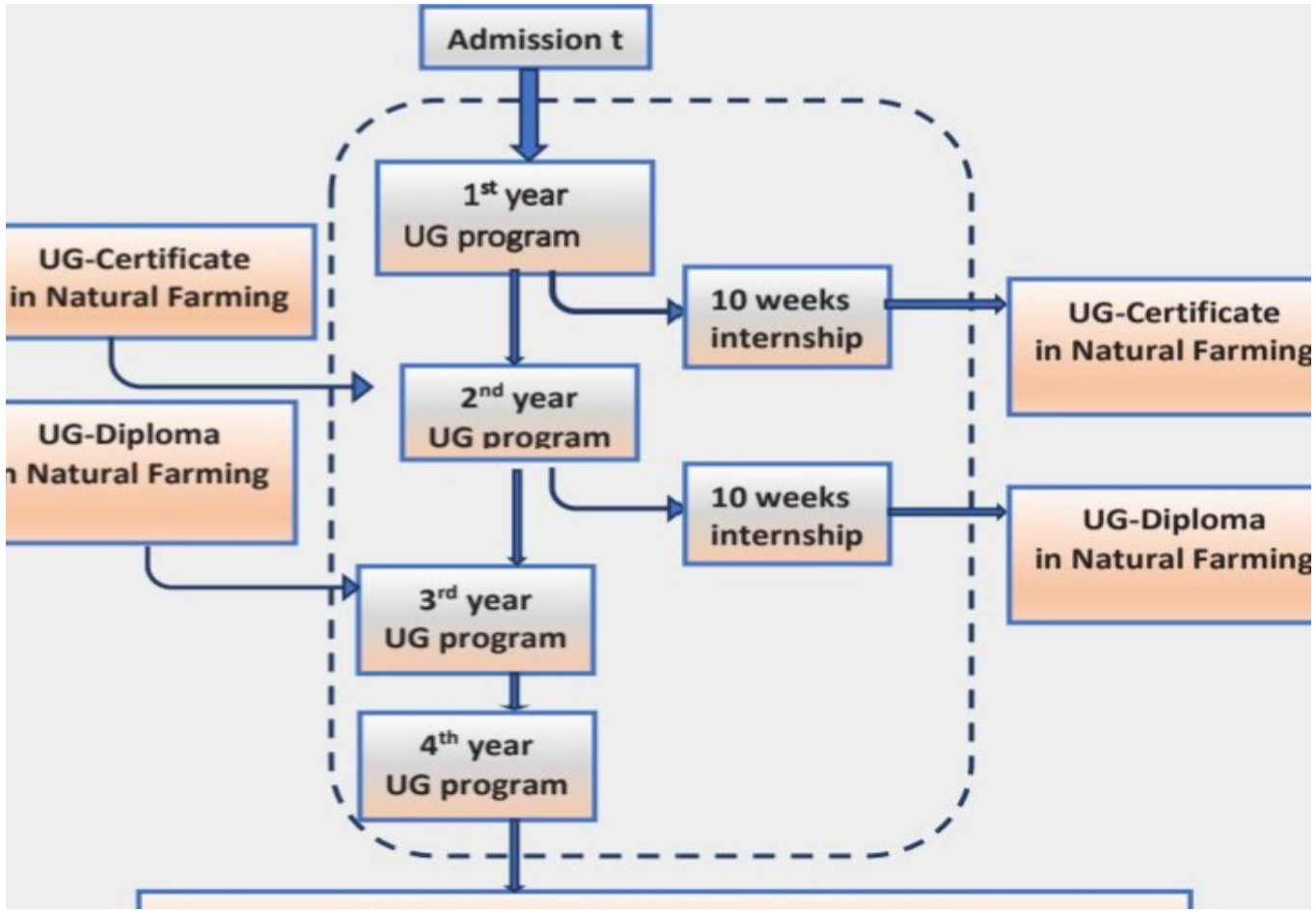
HIGHLIGHTS

- An institution IANS, DDUGU, Gorakhpur is at liberty, and in fact it should work in partnership with capable organizations/ companies/ NGOs/ progressive entrepreneurs for running various SEC programs. In such cases, while the parent institution will control admission process, develop the content and delivery of the program and monitor the learning and skill development by students and the evaluation can be done jointly by the collaborating partners.
- The second year has been designed with the practical exposure on natural farming as well as some part of fundamental courses related to agricultural sciences with adequate theory and practical components, enabling the student to get acquainted with the basic principles and applications of natural farming. After satisfactory completion of the courses of 2nd year and subsequent completion of 10 credits (10 weeks) of industry/ institute training/ internship, the student will become eligible for the award of UG-Diploma in Natural Farming on exit. The students continuing the study further, would not have to attend the internship after 2nd year.
- The third year and fourth year courses have been designed to impart in depth details including specialized knowledge to the students in the major disciplines. In order to inculcate the moral and experiential habits in students to, research methodology and ethics were also added besides Student READY (Rural and Entrepreneurship Awareness Development yojana) include Rural Agricultural Work Experience program/experiential learning/internship and student projects.
- There will be adequate choice of electives/ specialized courses for the students in the final year. The Universities will have flexibility to include more courses as electives depending on specific needs. The objective is to enable the student to acquire deeper understanding in any particular field. As regards individual course contents, universities/colleges may modify the content to the extent of 20 percent whenever necessary as per the regional demands and needs. There will be choice of elective/ specialization in final year for students.

HIGHLIGHTS

- Students will also be given freedom to opt 20 credits of non-credit courses from MOOC or any other portal in on line mode to groom up her/his passion and add on the knowledge components based on her/ his choice. There will also be flexibility to complete these non -credit elective courses of 20 credits throughout the span of the degree program.
- The online courses can be from any field such as Agriculture, Basic Sciences, Humanities, Psychology, Anthropology, Economics, Business Management, Languages including Foreign Languages, Communication Skills/ Music, etc. "Ihe objective is to allow the students to groom their passion or strengthen their knowledge and competency in any field beyond prescribed courses. These online courses will be non-gradial as separate certificates would be issued by institutes offering the courses. However, the university/ institute will keep a record of such courses registered and completed by each student and will indicate the title of the (successfully completed) courses in final transcript issued to the student.
- In eighth semester of the degree Program, students will be offered 20 credits from Student READY Programs including RAWE/Project work/industrial attachment experiential learning, internship or any other option mentioned by the parent institute. The project work may be Natural Farming-based project/research based project or Industry based project depending on the interest of the students.

Entry and Exit Options



Semester Wise Course Distribution

First Year Semester I

Sr. Number	Course Title	Credit Hours
1	Deeksharambh (Induction cum Foundation course)	2(0+2)CNC
2	Indian Heritage of Natural Farming	3(2+1)
3	Principles and Practices of Natural Farming	4(3+1)
4	Bhumi Suposhan (Soil Nutrition)	3(2+1)
5	Farming based Livelihood System	3(2+1)
6	Communication Skills	3(2+1)
7	Skill Enhancement Course	2(0+2)
8	Skill Enhancement Course	2(0+2)
9	National Services Scheme/NCC/NSo	1(0+1)
10	Physical Education, First Aid and yoga Practice	CNC
	Total Credits	21(10+11)

Semester II

Sr. Number	Course Title	Credit Hours
1	Cow- based Natural Farming	3(1+2)
2	Soil Biology in Natural Farming	3(2+1)
3	Integration of Horticulture Crops under Natural Farming	2(1+1)
4	Insect Ecology, Nematodes and Pest Management	3(2+1)
5	Entrepreneurship Development and Business Management	3(2+1)
6	Personality Development	2(1+1)
7	Skill Enhancement Course (SECIII)	2(0+2)
8	Skill Enhancement Course(SEC-IV)	2(0+2)
9	NCC-II, NSS-II	1(0+1)
	Total Credits	**21(9+12)**

Second Year Semester III

Sr. Number	Course Title	Credit Hours
1.	Crop Management in Natural Farming	3(2+1)
2.	Production of Bio-inputs	2(1+1)
3.	Farm Power and Machinery	3(1+2)
4.	Livestock and Poultry Production	3(2+1)
5.	Management of Plant Diseases	2(1+1)
6.	Biodiversity Conservation	2(1+1)
7.	Physical Education, First Aid and Yoga Practice	1(0+1)
8.	Skill Enhancement Course (SEC-V)	2(0+2)
9.	Environmental Studies and Disaster Management	3(2+1)
	Total Credit	**21(11+10)**

Semester IV

Sr. Number	Course Title	Credit Hours
1.	Elements, Characteristics and Design of Natural Farming Systems	3(2+1)
2.	Improvement of Traditional and Under-utilized Crops	3(2+1)
3.	Agro-Ecosystem Analysis for Natural Farming	2(1+1)
4.	Beneficial Insects	3(2+1)
5.	Quantification and Valuation of Ecosystem Services	3(2+1)
6.	Agricultural Marketing and Trade	3(2+1)
7.	Agricultural Informatics and Artificial Intelligence	2(2+1)
8.	Skill Enhancement Course (SEC- VI)	2(0+2)
	Total Credit	**22(13+9)**

Third Year Semester V

Sr. Number	Course Title	Credit Hours
1.	Post-Harvest Management –I	3(2+1)
2.	Bio- resources and Agricultural Waste Management	3(2+1)
3.	Medicinal and Aromatic Plants	2(1+1)
4.	Seed Production Technology	3(2+1)
5.	Value Chain Management in Natural Farming	2(1+1)
6.	Renewable Energy Sources	2(1+1)
7.	Natural Farming and Human Health	2(1+1)
8.	Fundamentals of Entomology	3(2+1)
9.	Educational Tour*	CNC
	Total	**20(12+8)**

Semester VI

Sr. Number	Course Title	Credit Hours
1.	Indian Traditional Knowledge	3(2+1)
2.	Post-harvest Management-II	3(2+1)
3.	Standards and Certification for Natural Farming	3(2+1)
4.	Aqua-based Natural Farming	3(2+1)
5.	Marketing of Natural Farming Produce	3(2+1)
6.	Agronomic Practices	2(1+1)
7.	Intellectual Property Rights	1(1+0)
8.	Principles of Agricultural Economics and Farm Management	2(2+0)
	Total	**20(14+6)**

Fourth Year Semester VII

Sr. Number	Course Title	Credit Hours
1.	Community Mobilization for Natural Farming	2(1+1)
2.	Water Management	2(1+1)
3.	Research Methodology and Ethics	2(1+1)
4.	Weather Forecasting	3(2+1)
5.	Basic and Applied Agricultural Statistics	3(2+1)
6.	Elective Course-I	3(2+1)
7.	Elective Course-II	3(2+1)
8.	Elective Course-III	3(2+1)
	Total Credit	**21(13+8)**

Semester VIII

Course Number	Course Title	Credit Hours
Project Work/ Experi ential Learning/Int ernship	**For Student opting for 4 Year B.Sc. Ag. (Hons.) Natural Farming Degree** Student READy (RAWE/Industrial Attachment/ Experiential Learning/Hands on Training/Project Work)/ Internship **For Student opting for 4 Year B.Sc. Ag (Hons.) Natural Farming (With Research)** Research work	20 credits (Minimum 20 weeks) 20 credits
	Total	**20(0+20)**

Vidyaramh: At the start of 1st Semester, there will be a three weeks Foundation Course (FC) with 0+4 credits (Non-gradial), common to all the students

* Educational (Study) Tour: It will be compulsory non-gradial of 2 credit for 10-12 days during or after 5th semester.

CNC – Compulsory Non-Credit Course.

Semester wise Credit Allocation (NF)

Semester	Total Credits	Courses (Major+ Minor)	MultiDisciplinary Course (MDC)	Value Added Cources	Ability Enhancement Course (AEC)	Skill Enhancement Course (SEC)	Internship/ Project/ Student READY	Non-Gradial	Online Courses/ MOOC
I	21	10							
II									
III									
IV									
V									
VI									
VII									
VIII									
Total									

Summery of credit distributions among different categories of courses

Semester	Total Credit	Skil enhancement courses	Basic Courses Major/ Minor	Common Courses	NCC/NSS/ Seminar/ Educational Tour	Student Ready/ Intern ship	Elective Courses	Online Courses
1st Semester	21+ FC*	4	13 Thery+ Practical	#3 Farming based Livelihood Systems	01		-	
2nd Semester	21	4	11Thery+ Practical	#2+3 Introductory English Communication Skil & Personality Development (2 Credit environmental Studies & Disasters Management	01	Credit *10 Weeds +10 Extra		
3rd Semester	20	4	13 Thery+ Practical	# 3 Agri Informatics & Data base Management	-	-		
4th Semester	22	4	14 Thery+ Practical	#3 Entrepreneurship Development & Business Management	01 Seminar	Credit *10 Weeds +10 Extra Time Period after 2nd year		

Summery of credit distributions among different categories of courses

Semester	Total Credit	Skil enhancement courses	Basic Courses Major/ Minor	Common Courses	NCC/NSS/ Seminar/ Educational Tour	Student Ready/ Internship	Elective Courses	Online Courses
5th Semester	20	-	20	-	2 Credit Educational Ture / NCC	-		
6th Semester	20	-	18	#1 Intellectual Properties Rights	Seminar	-		
7th Semester	21	-	12	-	-	-	Credits: May Select three courses each of 3 Credits from elective courses	
8th Semester	20	-	-	-	-	20	-	
All Semesters	165+20	16	101	15	4	20	09	20

- Vidhyarambh FC*= Foundation Course; At the start of 1st Semester there will be three weeks non gradial foundational course common to all students entering certificate course of all degree Programmes.
- # Comen Courses including both ability enhancement and value-added courses.
- Internship** of 10 weeks 10 credits, after the 2nd and 4th semester this internship will be done by the students after completion of respective semester and before of the certificate/ diploma in additional period.
- A student will make his own planning and execution of online courses with intimation to Director IANS.
- Education (Study) Toure: Compulsory non credit course for ten to twelve days.

Abstract	
Core courses :	101 credits
Skill Enhancement courses :	16 credits
Common courses :	15 credits
NCC/NSS/ yoga/Educational Tour/ : Seminar	4 credits
Elective courses :	9 credits
Project work and Internship work :	20 credits
MooC courses (non-gradial) :	20 credits
Grand Total :	185 credits

Abstract of Credit Hours

Semester	Credit hours	Contact hours
1st Semester	10+11=21	32
2nd Semester	9+12=21	33
3rd Semester	9+11=20	31
4th Semester	11+11=22	33
5th Semester	12+8=20	28
6th Semester	14+6=20	26
7th Semester	13+8=21	29
8th Semester	0+20=20	40
Total Credit hours	78+87=165	252
*online MooC courses (Non-Gradial)	20*	Choice based
Grand Total	165+20*=185	

Break Up of Category of Courses

Year	Semester	Credits	Breakup of categories of courses				
			Core courses (Major/ Minor/ Common)	**Skill Enhancement Courses/ Elective courses****	**NSS/ NCC/ Yoga/Se Minar/ Phy. Education**	**Internship and project work**	**#Online course (MOOC)**
I	1st	21	16	**4	1	-	Flexibility/ choice in selecting courses and completing in whole duration of the degree Program
	2nd	21	16	4	1	-	
II	3rd	20	16	4	-	-	
	4th	22	17	4	1	-	
III	5th	20	20	-	-	-	
	6th	20	19	-	1	-	
IV	7th	21	12	9	-	-	
	8th	20*	-	-	-	20	
Total		**165**	**116**	**25**	**4**	**20**	**20 (Non Gradial)**
Grand Total		**165+20* = 185**					

Student will take in online mode with approval from Dean/Competent Authority. * Student READy/Internship of 20 credits in 8th Semester, ** In Natural Farming discipline, the skill component need to be strengthened

List of Basic Supporting Courses

S. No.	Course Tittle	Credit Hours
1.	Vidyarambh (Foundation Course) Physical Education, First Aid and yoga Practice	CNC(0+4) CNC(0+1)
2.	NCC/NSS/NSo/yoga/Educational Tour/Seminar (Offered in first IV semesters)	2(0+2)
3.	Farming Based Livelihood Systems	3(2+1)
4.	Agri- Informatics and Data Base Management	(2+1)
5.	Environmental Studies and Disaster Management	3(2+1)
6.	Entrepreneurship Development and Business Communication	3(2+1)
7.	Research Methodology and Ethics	2(1+1)
8.	Intellectual Property Rights	1(1+0)

List of Courses with Credits as per Categories for B.Sc. Ag. (Hons.) Natural Farming

Course Category	Course Title	Credit Hours
Foundation course	Vidyarambh (3 weeks)	0+4 (NC)
Common courses	Farming based Livelihood System	3(2+1)
	Introductory English, Communication Skills and Personality Development	2(1+1)
	Environmental Studies and Disaster Management	3(2+1)
	Agri- Informatics and Database Management	3(2+1)
	Entrepreneurship Development and Business Communication	3(2+1)
	Intellectual Property Rights	1(1+0)
	Total credits	15(10+5)

Core courses	Indian Heritage of Natural Farming Principles and Practices of Natural Farming	3(2+1) 4(2+2)
	Bhumi Suposhan (Soil Nutrition)	3(2+1)
	Biodiversity and Conservation	3(2+1)
	Cow- based Natural Farming	3(1+2)
	Soil Biology in Natural Farming	3(2+1)
	Integration of Horticulture Crops under Natural Farming	2(1+1)
	Insect Ecology, Nematodes and Pest Management	3(2+1)

Course Category	Course Title	Credit Hours
Elective courses	Elective Course-I	3(2+1)
	Elective Course-II	3(2+1)
	Elective Course-III	3(2+1)
	Total credits	9 (6+3)
	Skill Enhancement Course I	2(0+2)

Course Category	Course Title	Credit Hours
Core courses	Crop Management in Natural Farming	3(2+1)
	Production of Bio-inputs	2(1+1)
	Farm Power and Machinery	3(1+2)
	Livestock and Poultry Production	3(2+1)
	Management of Plant Diseases	2(1+1)
	Elements, Characteristics and Design of Natural Farming Systems	3(2+1)
	Improvement of Traditional and Under-utilized Crops	3(2+1)
	Agro-Ecosystem Analysis for Natural Farming	2(1+1)
	Beneficial Insects	3(2+1)
	Quantification and Valuation of Ecosystem Services	3(2+1)
	Post-Harvest Management –I	3(2+1)
	Bio- resources and Agricultural Waste Management	3(2+1)
	Medicinal and Aromatic Plants	2(1+1)
	Seed Production Technology	3(2+1)
	Value Chain Management in Natural Farming	2(1+1)
	Renewable Energy Sources	2(1+1)
	Natural Farming and Human Health	2(1+1)
	Indian Traditional Knowledge	3(2+1)
	Post-harvest Management-II	3(2+1)
	Standards and Certification for Natural Farming	2(2+0)
	Aqua-based Natural Farming	3(2+1)
	Marketing of Natural Farming Produce	3(2+1)
	Agronomic Practices	2(1+1)
	Community Mobilization for Natural Farming	2 (1+1)
	Water Management	2 (1+1)
	Research Methodology and Ethics	2 (1+1)
	Weather Forecasting	3 (2+1)
	Fundamentals of Entomology	3(2+1)
	Principles of Agricultural Economics and Farm Management	2(2+0)
	Basic and Applied Agricultural Statistics	3(2+1)

Course Category	Course Title	Credit Hours
Skill Enhancement Courses	Skill Enhancement Course II	2(0+2)
	Skill Enhancement Course III	2(0+2)
	Skill Enhancement Course IV	2(0+2)
	Skill Enhancement Course V	2(0+2)
	Skill Enhancement Course VI	2(0+2)
	Skill Enhancement Course VII	2(0+2)
	Skill Enhancement Course VIII	2(0+2)
	Total credits	16 (0+16)
Project Work/ Internship	Student READy (RAWE/ Industrial Attachment/ Experiential Learning/Hand on Training/Project Work)/ Internship	20 (0+20)
	Experiential Learning	8
	RAWE	6
	IPT/Industrial Attachment	4
	Student Project	2
	Total credits	20 (0+20)
Other courses	NSS/NCC/NSo/Phy Edn./ yoga - I	1 (0+1)
	NSS/NCC/NSo/Phy Edn./ yoga - II	1 (0+1)
	Educational Tour	NG
	Seminar-I	1 (0+1)
	Seminar-II (Case studies)	1 (0+1)
	Total credits	6 (0+6)
	Total for offline courses	165 (78+87)
On-line courses	online course	20 (0+20)
	Total for online courses	20 (0+20)
	GRAND TOTAL	185 (78+107)

*NG - Non-Gradial

Detailed Syllabi

Semester I

Deeksharambh (lnduction cum Foundation Course) 2 (0+2) (Non-Gradial)

The details of activities will be decided by the host universities. The structure should include, but not restricted to:

(i) Discussions on operational framework of academic process in university, as well as interactions with academic and research managers of the University.

(ii) Interaction with alumni, business leaders, scientists and prospective employers

(iii) Group activities to identify the strength and weakness of students (with expert advice for their improvement), streaming of the students as per their aptitude for research entrepreneurship or others through psychometric test as well as to create a platform for students to learn from each others life experiences.

(iv) To identify the students having difficulty in mathematics and enrolling them for I non-gradial remedial course on mathematics of one credit.

(v) Activities to enhance cultural integration of students from different backgrounds.

(vi) Classes on personality development (instilling life and social skills, social awareness, ethicsand values, team work, leadership, Knowing and Being etc.) and communication skills.

Indian Heritage of Natural Farming 3 (2+1)

Objectives

1. To impart knowledge on the heritage of natural farming in ancient India and various techniques of Vedic Krishi.
2. To develop skills on Vedic Krishi practices for practicing natural farming.

Theory

The heritage of natural farming, Pioneers and scholars of natural farming and their contribution to the heritage of natural farming, Ancient methods of livestock management, health, nutrition, soil fertility and plant protection, Description on ancient methods of rainfall prediction, Importance of seed and their collection, storage, removal of weed seeds and seed uniformity, Discussions on important rare documents like Krishi-Parashar, Sitadhyaksha chapter in Kautilya Arthashastra (Acharya Kautilya-321 BC), Kashyapiya Krishi Sukti (by sage Kashyap-c.800 CE), Vriksha yurveda (by Vaidya Surapala), Upavanavinod (Nalini, 2011), Vishwavallabha (by Chakrapani Mishra-1577 CE), Brihat Samhita (by Varahamihi- 600 AD), Lokopakara (1000-year-old manuscript), Nuskha Dar FanniFalahat-The Art of Agriculture(by Prince Dara Shikoh-1650 CE), Krishi Gita-Agricultural Versus (15th century by Vidwan C Govinda Warriar), etc. Measurement ofwealth on natural resources, 'Gaudhan' (Cows), 'Ashwa-dhan' (Horses), 'Gaj-dhan' (Elephants) etc. as all different forms of wealth. Vidya-dhan'. The most important wealth 'dhanya' or rice/crops, Mode ofthe transactions in the society through 'dhanya'. Broad tradition of 'natural farming', propounded by advocates such as Dr Y.L Nene, (Asian Agri History Foundation), Narayana Reddy (Karnataka), Shripad Dabholkar (Maharashtra), G Nammalvar (Tamil Nadu), Partap C Aggarwal (Madhya Pradesh) and Bhaskar Hiroji Save (Gujarat). Rich heritage of plant protection in natural farming, Identification of disorders; Vrikshayurveda describes various plant diseases based on Tridosha (Vata, Pitta, Kapha) Siddhanta of Ayurveda along with symptoms and remedial measures. Plant protection practices, Kunapajala, Herbal Kunapajala and other indigenous liquid concoctions

Indian Heritage of Natural Farming 3 (2+1)

Practical

- Visit of natural farms to understand the concept, components and management practices.
- Identification of plant and animal-based products used in Vrikshayurveda for plant nutrition and crop protection.
- Preparation of Vrikshayurveda based on-farm inputs for nutrient, pest and disease management.
- Documentation of vrikshayuveda technology knowledge-base for resource conservation, soil health and plant protection.

Principles and Practices of Natural Farming 4 (3+1)

Objectives

1. To make the students aware about the agro-ecology, concept and principles.
3. To impart knowledge on various principles and practices in natural farming.
4. To impart knowledge about different methods of natural farming.

Theory

Natural farming: Concept and importance. Modern agriculture and need for new strategies, origin of concept of natural farming, Agroecology: Concept and Principles, history of development in agro-ecology, Natural farming: principles, philosophy and components, Merits and demerits of natural farming, Principles of Fukuoka and other methods of natural farming, Practices of Natural Farming - case studies, Types of natural farming systems (Community based natural farming, Palekar's Natural Farming, Bio-dynamic, homa farming, natu- eco farming, rishi Krishi, panchagavvya Krishi, yogic farming etc., Scope of natural farming in modern time, Laws of nature, Indian philosophy /Bhartiya Chintan of Pancha Mahabhoot and their role in human life and development, Important management practices for crops, fruits, vegetables and animals in natural farming, comparison of conventional, organic and natural farming production systems, Scientific aspects of ecological and natural farming for supporting food, nutritional and livelihood security food sovereignty, livelihood of farmers and rural communities, biodiversity and conservation of biodiversity, Sustainable soil health and clean water, Ecological pest protection, Climate resilient food production, Indigenous farming systems prevalent in India and other countries. Government schemes supporting natural farming (PKVy, BPKP, National Mission on Natural Farming), StartUps in chemical free agriculture, Digital India.

Principles and Practices of Natural Farming 4 (3+1)

Practical

- Identification of indigenous crops and practices,
- Study of different farming systems and their characterization,
- Students' visit to natural farms to study the various components and practices of natural farming,
- Development of natural farm inventory,
- Analysis of ancient Indian and modern natural farming,
- Bio- dynamics,
- Homa farming, natu-eco farming, rishi Krishi, panchagavvya Krishi, Yogic farming and practical field application of these practices,
- Design of natural farm based on locally available resources, Preparation of on- farm inputs for nutrients, pest and disease management.
- Documentation of indigenous technology knowledge (ITK) for nutrient, insect, pest disease and weed management,
- Development of documentary on management of natural farm.

Bhumi Suposhan (Soil Nutrition) 3 (2+1)

Objectives

1. To impart knowledge on various properties of soils, soil evaluation procedures, factors of soil fertility and management of biological processes through natural farming practices
2. To develop skills among students on soil management practices under natural farming

Theory

Soil as a natural medium for plant growth, Status of soil degradation; Weathering of rocks and minerals, Soil genesis and classification; Concept of essential, beneficial, functional and trace elements in soils; Role of essential nutrients in plants, their deficiency and toxicity symptoms and diagnosis, Nutrient movement in soils and mechanisms of nutrients uptake by plants, Cycling of nutrients in soils (C, N, P, S, secondary and micronutrients), Importance of physical, chemical, and biological properties of soil, Factors affecting soil fertility, Methods of soil fertility evaluation: physical, chemical and biological, origin, formation, extent of distribution and characteristics of acid soils, acid sulphate soils, alkali soils, saline soils and their management, Formation of soil organic matter and its management in agroecosystem, carbon sequestration, soil carbon and nitrogen management indices, Soil quality and health, soil quality indicators, characteristics of a healthy soil, Principles of supporting healthy rhizosphere interactions for improved soil fertility, Best management practices (BMPs) for promoting soil health and monitoring of soil-health: minimum tillage, mulching, organic residue incorporation, cover crops, green manuring, crop rotation, conducive environment for beneficial microbes, mixed cropping, intercropping, composts, FYM, biochar, bio- culture formulations, Sing Khad, (Horn manure), Kunapajal, Samadhi Khada, biogas manures (liquid and solid), etc., Principles and methods of production of various organic manures, Sources of contamination in soil and reclamation by bioremediation processes (Phytoremediation, phytoextraction, phytostabilization and microbial bioremediation), Ancient classification of soil based on soil fertility, irrigation, physical characteristics, and crop suitability; Ancient practices for soil enrichment under natural farming, Traditional festivals of the region with respect to soil health, Schemes promoting soil health in India.

Bhumi Suposhan (Soil Nutrition) 3 (2+1)

Practical

- Soil sampling methods,
- processing, and storage, Determination of soil texture (feel method, hydrometer method and international pipette method),
- Determination of soil bulk density, particle density, porosity, and water holding capacity,
- Determination of soil pH and EC, Determination of soil organic carbon,
- Determination of available nitrogen,
- Determination of available phosphorus,
- Determination of available potassium,
- Determination of available secondary plant nutrients (Ca, Mg, S,),
- Determination of available micronutrients (Zn, Fe, Cu, Mn, B, MO),
- Plant sample digestion procedure,
- Analysis of important nutrient elements in plant,
- Studies on important ITKS for selection of crop-soil combination,
- Quantification of carbon-neutral farming practices.

Farming based Livelihood Systems 3 (2+1)

Objective

To disseminate the knowledge and skill how farming systems can be a source of livelihood

Theory

Concept of farming system and integrated farming system- Definition, goal and advantages, Classification of farming systems, Components of farming systems- Crops and cropping systems, Livestock and poultry, Fishery and Secondary enterprises, Integration of components in farming system, Factors affecting integration of components in IFS, Selection of crops and cropping systems for various enterprises, Feasibility of different cropping systems for different agro- climatic zones, Horticultural crops and cropping systems feasible for different farming systems, Livestock and poultry as a component of integrated farming systems, Study of different livestock components feasible for IFS (Dairy, Piggery, goatry, poultry, duckry etc.), Aqua culture as a component of IFS, Duck/Poultry cum Fish, Dairy cum fish, Piggery cum fish etc. Problems, profitability and prospects, Sericulture as a component of IFS: Scope and potentiality of sericulture for small and marginal farmers of states, some suitable module of IFS for different agro-ecological situations, Commercial livestock production- problems and prospects, Design for livestock shed and their maintenance, Importance of feed processing unit for livestock production, Silage and hay production, Marketing, Concept of sustainable farming system and their indicators.

Farming based Livelihood Systems 3 (2+1)

Practical

- Study of IFS models for different agroclimatic zones,
- Preparation of IFS model in groups,
- Recycling of organic waste in IFS,
- Study of production and profitability of IFS,
- Market survey for IFS products,
- Preparation of organic manures in IFS,
- Study of crops and cropping systems for diversification in IFS,
- Visit to farmer's field for feasibility study for scientific intervention in IFS,
- Study of Integrated Organic Farming Systems.

Communication Skills 2 (1+1)

Objective

To acquire competence in oral, written and non-verbal communication, develop strong personal and professional communication and demonstrate positive group communication.

Theory

- Communication Process: The magic of effective communication; Building self-esteem and overcoming fears; Concept, nature and significance of communication process; Meaning, types and models of communication; Verbal and non-verbal communication; Linguistic and non-linguistic barriers to communication and reasons behind communication gap/ miscommunication.
- Basic Communication Skills: Listening, Speaking, Reading and Writing Skills; Precis writing/ Abstracting/Summarizing; Style of technical communication Curriculum vitae/resume writing; Innovative methods to enhance vocabulary, analogy questions.
- Structural and Functional Grammar: Sentence structure, modifiers, connecting words and verbals; phrases and clauses; Case: subjective case, possessive case; objective case; Correct usage of nouns, pronouns and antecedents, adjectives, adverbs and articles; Agreement of verb with the subject: tense, mood, voice; Writing effective sentences; Basic sentence faults.

Communication Skills 2 (1+1)

Practical

- Listening and note taking;
- Writing skills: precis writing, summarizing and abstracting;
- Reading and comprehension (written and oral) of general and technical articles;
- Micro-presentations and Impromptu Presentations: Feedback on presentations;
- Stage manners: grooming, body language, voice modulation, speed;
- Group discussions; Public speaking exercises;
- Vocabulary building exercises; Interview Techniques; organization of events.

Skill Enhancement Courses 4 (0+4)

Objective

- To enhance the employment potential and to set up small business enterprises skills of volunteers, a list of 12 to 15 vocational skills will be drawn up based on the local conditions and opportunities. Each volunteer will have the option to select two skill-areas out of this list of issues related environment.
- Environmental conservation, enrichment and sustainability, climatic change, natural resource management (rainwater harvesting, energy conservation, forestation, waste land development and soil conservations) and waste management. Disaster management. Introduction and classification of disaster, rehabilitation, and management after disaster; role of NSS volunteers in disaster management.
- Entrepreneurship development. Definition, meaning and quality of entrepreneur; steps in opening of an enterprise and role of financial and support service institution.
- Formulation of production-oriented project. Planning, implementation, management, and impact assessment of project. Documentation and data reporting. Collection and analysis of data, documentation, and dissemination of project reports

Skill Enhancement Courses 4 (0+4)

Two Courses each of (0+2) credits of skill enhancement courses will be offered from indicated list of skill enhancement courses.

National Cadet Corps (NCC-I) 1 (0+1)

- Aims, objectives, organization of NCC and NCC song. DG's cardinals of discipline.
- Drill- aim, general words of command, attention, stands at ease, stand easy and turning.
- Saluting at the halt, getting on parade, dismissing, and falling out.
- Marching, length of pace, and time of marching in quick/slow time and halt. Side pace, pace forward and to the rear. Turning on the march and wheeling. Saluting on the march.
- Marking time, forward march, and halt. Changing step, formation of squad and squad drill.
- Command and control, organization, badges of rank, honors, and awards
- Nation Building- cultural heritage, religions, traditions, and customs of India. National integration. Values and ethics, perception, communication, motivation, decision making, discipline and duties of good citizens. Leadership traits, types of leadership. Character/ personality development. Civil defense organization, types of emergencies, firefighting, protection. Maintenance of essential services, disaster management, aid during development projects.
- Basics ofsocial service, weaker sections of society and their needs, NGO's and their contribution, contribution of youth towards social welfare and family planning.
- Structure and function of human body, diet and exercise, hygiene and sanitation. Preventable diseases including AIDS, safe blood donation, first aid, physical and mental health. Adventure activities. Basic principles of ecology, environmental conservation, pollution and its control.

Skill Enhancement Courses 4 (0+4)

National Service Scheme (NSS-I) 1 (0+1)

All the activities related to the National Service Scheme are distributed under four different courses viz., NSS- I, NSS- Il, NSS- Ill and NSS- IV each having one credit load. The entire four courses should be offered continuously for two years.

A student enrolled in NSS course should put in at least 60 hours of social work in different activities in a semester other than five regular one-day camp in a year and one special camp for duration of 7 days at any semester . Activities directed by the Central and State Government have to be performed by all the volunteers of NSS as per direction.

Semester l: Introduction and Basic Components of NSS

- Orientation: history, objectives, principles, symbol, badge; regular programs under NSS
- organizational structure of NSS, Code of conduct for NSS volunteers, points to be considered by NSS volunteers' awareness about health.
- NSS program activities. Concept of regular activities, special camping, day camps, basis of adoption of village/slums, conducting survey, analyzing guiding financial patterns of scheme, youth program/ schemes of GOI, coordination with different agencies and maintenance of diary. Understanding youth. Definition, profile, categories, issues and challenges of youth; and opportunities for youth who is agent of the social change.
- Community mobilization. Mapping of community stakeholders, designing the message as per problems and their culture; identifying methods of mobilization involving youth-adult partnership. Social harmony and national integration
- Indian history and culture, role of youth in nation building, conflict resolution and peacebuilding. Volunteerism and shramdaan. Indian tradition of volunteerism, its need, importance, motivation, and constraints; shaman as part of volunteerism
- Citizenship, constitution, and human rights. Basic features of constitution of India, fundamental rights and duties, human rights, consumer awareness and rights and rights to information. Family and society. Concept of family, community (PRIS and other community based organizations) and society.

Skill Enhancement Courses 4 (0+4)

Physical Education, First Aid, Yoga Practices and Meditation 1(0+1)

Objectives

1. To make the students aware about Physical Education, First Aid and Yoga Practices
2. To disseminate the knowledge and skill how to perform physical training, perform first aid and increase stamina and general wellbeing through yoga.

Practical

Physical education; Training and Coaching - Meaning and Concept; Methods of Training; aerobic and aerobic exercises; Calisthenics, weight training, circuit training, interval training, Fartlek training; Effects of Exercise on Muscular, Respiratory, Circulatory &Digestive systems; Balanced Diet and Nutrition: Effects of Diet on Performance; Physiological changes due to ageing and role of regular exercise on ageing process; Personality, its dimensions and types; Role of sports in personality development; Motivation and Achievements in Sports; Learning and Theories of learning; Adolescent Problems and its Management; Posture; Postural Deformities; Exercises for good posture.

Yoga; History of Yoga, Types of Yoga, Introduction to Yoga,

- Asanas (Definition and Importance) Padmasan, Gaumukhasan, Asanas (Definition and Importance) Padmasan, Gaumukhasan, Bhadrasan, Vajrajasan, Shashankasan, Pashchimotasan, Ushtrasan, Tadasan, Padhastasan, Ardhchandrasan, Bhujangasan, Utanpadasan, Sarvangasan, Parvatasan, Patangasan, Shishupalanasan — left leg-right leg, Pavanmuktasan, Halasan, Sarpasan, Ardhdhanurasan, Sawasan Suryanamskar Pranayama (Definition and Importance) Ornkar, Suryabhedan, Chandrabhedan, AnulomVilom, Shitali, Shitkari, Bhastrika, Bhramari, Meditation (Definition and Importance), Yogic Kriyas (Kapalbhati), Tratak,

Skill Enhancement Courses 4 (0+4)

Physical Education, First Aid, Yoga Practices and Meditation 1(0+1)

- **Mudras (Definition and Importance)**

Gyanmudra, Dhyanmudra, Vayumudra, Akashmudra, Pruthvimudra, Shunyamudra, Suryamudra, Varunmudra, Pranmudra, Apanmudra, Vyanmudra, Uddanmudra

- **Role of yoga in sports**
- **Teaching of Asanas** — demonstration, practice, correction and practice.
- **History of sports and ancient games**, Governance of sports in India; Important national sporting events; Awards in Sports; History, latest rules, measurements Of playfield, specifications of equipment, skill, technique, style and coaching of major games (Cricket, football, table Tennis, Badminton, Volleyball, Basketball, Kabaddi and Kho-Kho) and Athletics
- **Need and requirement of first aid.**

First Aid equipment and upkeep. First AID Techniques, first aid related with respiratory system. First aid related with Heart, Blood and Circulation. First aid related with Wounds and Injuries. First aid related with Bones, Joints Muscle related injuries. First aid related with Nervous system and Unconsciousness. First aid related with Gastrointestinal Tract. First aid related with Skin, Burns. First aid related with Poisoning. First aid related with Bites and Stings. First aid related with Sense organs, Handling and transport of injured traumatized persons. Sports injuries and their treatments.

Semester II

Cow Based Natural Farming 3 (1+2)

Objectives

1. To acquaint the students about practicing natural farming involving cow and its relationship with farming activities
2. To impart skill on farming practices and technologies in cow based natural farming

Theory

Sociology of natural farming, Indian Livestock scenario, Indigenous breeds of cattle, Differentiating features of indigenous, crossbred and exotic cattle, Traditional livestock farming practices, Draft animal power in natural farming, Feeding cattle at various phases, Forage production, Hay making, Preventive health care practices, Bio efficacy of cow urine on crop production, Effect of different natural inputs on yield, quality and soil properties, Effects of cow urine and its major constituents on pasture properties, Cow based rearing systems: land, feed and water requirement of traditional farming method, Advantages of cow based natural farming methods, Economics of cultivation and comparison with currently adopted practices, Role of cow based farming system in utilising the available cattle (Desi cow) as valuable resource, Cost of cultivation under cow based natural farming, Composition and constituents of desi cow milk vs cross breeds; role of tryptophan precursor of serotonin in relaxing human mind and better sleep, role of A2 Milk in controlling blood sugar level. Sustainable agriculture and fodder business, Natural farming business and entrepreneurship development, Supply chain management, Application of bio gas in agriculture, Traditional therapies for various ailments, ITKS in cow based natural farming.

Cow Based Natural Farming 3 (1+2)

Practical

- Identification and characteristics of various cattle breeds,
- Fodder varieties and feed ingredients;
- Visit to dairy farm to study the various components and their utilization, Visit to gau-shala, Visit to an institute involved in cow based natural farming,
- Design and development of cow based natural farm,
- Identification of herbal plants used for livestock treatment,
- Ethno veterinary medicines,;
- Feed and fodder conservation practices-Silage and Hay making, Concentrate making techniques including ration balancing for cow based farms,
- Calf management under natural Farm,
- Value addition of milk from dairy, Grassland management, Value addition of by-products from cow based natural farm, Bio- inputs preparation for natural farms, Traditional practices followed in cow-based natural farms, Record keeping and economics of cow based natural farming, Preparation of inputs for cow based natural farming, Demonstration of renewable energy sources in cow based natural farms.

Soil Biology in Natural Farming 3 (2+1)

Objectives

- To impart knowledge on various aspects of soil biology, biological interactions and their roles in management of soil quality and productivity.
- To develop skills among students on isolation, purification, identification and use of soil beneficial microbes and fauna in natural farming system

Theory

Soil as a habitat for life, concept of soil biome and its components, Soil organic matter: Definition and its characterization, Biochemical processes of formation of soil organic matter, SOM as carbon source for soil biota, Decomposition processes, reaction order and its dynamics, Soil biodiversity and its methods of evaluation, Soil microbiota: its distribution and abundance, metabolism and physiology, classification of soil microorganisms and their importance in soil processes, Soil fauna (microfauna, mesofauna, macrofauna and megafauna): distribution, abundance, diversity and interactions; Factors affecting soil biodiversity, Significance of biota in soil development, Soil biological interactions: Functions and processes, Above-ground and below-ground biotic linkages, Rhizosphere, spermosphere, phyllosphere and their use in crop growth and development, Habitat adaptive fitness benefits to host crops in terms of tolerance to abiotic and biotic stresses, Biological nitrogen fixation: Symbiotic (Rhizobium-legume, Actinorhizal, Azolla-Anabaena), Associative (Azotobacter, Azospirillum, Gluconacetobacter, etc.), Free-living (BGA, Klebsiella, etc.), Microbiological methods of composting: various composting techniques, biofortification of various organic manures, In- situ decomposition techniques of crop residues using decomposers, Mechanisms of Biological Nitrogen Fixers (BNF), factor affecting BNF, Microbial mediated nutrient availability and exchange in soil, Nutrient mobilization, Arbuscular mycorrhizal symbiosis: its types, biochemical basis of interactions and benefits to host crops, Soil enzymes -characterization and mode of action, Agricultural management impacts on soil enzyme activities, Soil health, soil sickness, soil resistance concept, indicators and evaluation criteria, ITKS on soil health management through enriching soil microbiome.

Soil Biology in Natural Farming 3 (2+1)

Practical

- Hands on microscopy and examination of microbes,
- Microbiological media and methods of sterilization,
- Isolation and purification of microbial cultures,
- Enumeration of microbial population in soil- bacteria, fungi, actinomycetes,
- Isolation of Azospirillum, Azotobacter, BGA, Rhizobium,
- Determination of microbial biomass C and N, Assay for soil enzyme activities, Assay for carbon substrate utilization,
- Determination of earthworm population and biomass,
- Estimation of soil carbon active pool (co evolution, SMBC),
- Estimation Of different carbon fractions of soil, Estimation of soil carbon stock.

Integration of Horticultural Crops under Natural Farming 2 (1+1)

Objectives

1. To acquaint the students with package and practices for horticultural crops under natural farming.
2. To impart the practical training on horticultural crop production.

Theory

Importance and scope of horticulture crops in natural farming, Soil management in relation to nutrient and water uptake and their effect on soil environment, Intercrops- types and its role in nutrient management, Biological efficiency of cropping systems in horticulture, Planting of fruit plants in combination with annual crops — principles and strategies, Types of planting system for fruit plants in natural farming, Competitive and complimentary effect of root and shoot systems, Type of mulches - sod mulch and organic mulches for soil moisture and nutrients management, Irrigation — defined systems of irrigation for natural farming in fruit crops, Soil microbes and their role for sustainable horticultural crop production system, Canopy management, Training and pruning, Methods of training and training of important fruit trees, High density planting (HDP) and meadow orchards and their management, Rejuvenation of old orchards - top working and frame working, Integrated pest and disease management (IPDM), Integrated nutrient management (INM), Crop regulation in relation in fruit crops, Management of resources constraints in existing systems, Climate aberrations and mitigation measures of horticultural crops, Management of fruit plants of important tropical, sub-tropical and temperate fruit crops under natural farming, Mulching of fruit crops and Whapasa management, ITK for management of fruit plants under natural farming system.

Integration of Horticultural Crops under Natural Farming 2 (1+1)

Practical

- Selection of site for farm system design for establishment of natural farming of horticulture crops,
- Different planting systems and its layout — Triangular system, contour lines, double row system, rectangular and square, Preparation of pit and procedures for filling, Soil depth and slope for fruit tree plantation, Irrigation methods and application of water to fruit crops,
- Growing of intercrops ginger, turmeric, colocasia, cowpea, cabbage and green manuring,
- Filler crops — definition and their use in young orchard,
- Training and pruning methods of important tropical, sub-tropical and temperate fruit crops,
- Weed management in orchards,
- Use of plant-based products and microbes for plant protection of tropical, sub- tropical and temperate fruit crops,
- Mulching of fruit crops and Whapasa management

Insect Ecology, Nematodes and Pest Management 3 (2+1)

Objectives

1. To teach students the insect ecology and effect of biotic and abiotic factors on pests
2. To impart practical knowledge on various plant protection options under natural farming
3. To impart knowledge on plant parasitic nematodes and management options under natural farming

Theory

Insect Ecology - Introduction, environment and its components, Effect of abiotic factors — temperature, moisture, humidity, rainfall, light, atmospheric pressure and air currents, Effect of biotic factors — food competition, natural and environmental resistance, Categories of pests, Storage pests. Solutions for insect-pest control in natural farming: surveillance, perimeter weeding, conventional land preparation, host plant resistance, interaction of plant resistance traits and biocontrol agents, crop rotation and planting practices (date of sowing), intercropping/ mixed cropping, trap crops, border crops, sanitation, conserving natural enemies via habitat enrichment, Physical control strategies to reduce pest incidence, Mechanical control, Behavioral control (repellents and anti-feed ants plants, Biological control (Importation, Conservation, Augmentation; Parasitoids predators and Pathogens), Factors affecting success of biological control: Narrow host range/ Climatic adaptability/ Synchrony with host life cycle / High reproductive potential / Efficient search ability/ Short handling time / Survival at low host density, Botanicals (Plant extracts), Plant bioformulations: agniaster, brahmaster, neemaster, dasparni ark etc. Management of stored grain pests by non-chemical methods.

History and economic importance of nematodes: General characters of plant parasitic nematodes, their morphology, symptomatology and control of important plant parasitic nematodes of field and fruits by non-chemical methods, Use of various plant bioformulations like agniaster, brahmaster, neemaster, dasparni ark etc- for nematodes and pest management.

Practical

- Sampling techniques for estimation of insect-pests and nematodes populations,
- Collection of insect-pests, nematodes, natural enemies and their identification,
- Diversity of insect- pests, nematodes,
- Calculation of diversity indices,
- Preparation of different bioformulations and their application methods.

Entrepreneurship Development and Business Management: 3 (2+1)

Objectives

1. To provide student an insight into the concept and scope of entrepreneurship.
2. To expose the student to various aspects of establishment and management ofa small business unit.
3. To enable the student to develop financially viable agribusiness proposal.

Theory

Development of entrepreneurship, motivational factors, social factors, environmental factors, characteristics of entrepreneurs, entrepreneurial attributes/competencies. Concept, need for and importance of entrepreneurial development. Evolution of entrepreneurship, objectives of entrepreneurial activities, types of entrepreneurs, functions of entrepreneurs, importance of entrepreneurial development, and process of entrepreneurship development. Environment scanning and opportunity identification need for scanning—spotting of opportunity-scanning of environment— identification of product / service — starting a project; factors influencing sensing the opportunities. Infrastructure and support systems- good policies, schemes for entrepreneurship development; role of financial institutions, and other agencies in entrepreneurship development. Steps involved in functioning of an enterprise. Selection of the product / services, selection of form of ownership; registration, selection of site, capital sources, acquisition of manufacturing know how, packaging and distribution. Planning of an enterprise, project identification, selection, and formulation ofproject; project report preparation, Enterprise Management. Production management — product, levels of products, product mix, quality control, cost of production, production controls, Material management. Production management — raw material costing, inventory control. Personal management — manpower planning, labour turn over, wages / salaries. Financial management / accounting — funds, fixed capital and working capital, costing and pricing, long term planning and short-term planning, book keeping, journal, ledger, subsidiary books, annual financial statement, taxation. Marketing management- market, types, marketing assistance, market strategies. Crisis management- raw material, production, leadership, market, finance, natural etc.

Entrepreneurship Development and Business Management:3 (2+1)

Practical

- Visit to small scale industries/agro-industries,
- Interaction with successful entrepreneurs/ agric entrepreneurs.
- Visit to financial institutions and support agencies.
- Preparation of project proposal for funding by different agencies.

Personality Development 2 (1 +1)

Objective

To make students realize their potential strengths, cultivate their inter-personal skills and improve employability.

Theory

Personality Definition, Nature of personality, theories of personality and its types. The humanistic approach - Maslow's self-actualization theory, shaping of personality, determinants of personality, Myers-Briggs Typology Indicator, Locus of control and performance, Type A and Type B Behaviours, personality and Organizational Behaviour.

Foundations of individual behavior and factors influencing individual behavior, Models of individual behavior, Perception and attributes and factors affecting perception, Attribution theory and case studies on Perception and Attribution. Learning: Meaning and definition, theories and principles of learning, Learning and organizational behavior, Learning and training, learning feedback. Attitude and values, Intelligence- types of Intelligence, theories of intelligence, measurements of intelligence, factors influencing intelligence, intelligence and Organizational behavior, emotional intelligence. Motivation-theories and principles, Teamwork and group dynamics.

Practical

MBTI personality analysis, Learning Styles and Strategies, Motivational needs, Firo-B, Interpersonal Communication, Teamwork and team building, Group Dynamics, Win-win game, Conflict Management, Leadership styles, Case studies on Personality and Organizational Behavior.

Skill Enhancement Courses 4 (0+4)

Two Courses each of (0+2) credits of skill enhancement courses will be offered from indicated list of skill enhancement courses.

National Cadet Corps (NCC-II) 1 (0+1)

- Arms Drill- Attention, stand at ease, stand easy. Getting on parade. Dismissing and falling out. Ground/take up arms, examine arms. Shoulder from the order and vice-versa, present from the order and vice-versa. Saluting at the shoulder at the halt and on the march. Short/long trail from the order and vice- versa. Guard mounting, guard of honor, Platoon/Coy Drill
- Characteristics of rifle (.22/.303/SLR), ammunition, fire power, stripping, assembling, care, cleaning, and sight setting. Loading, cocking, and unloading. The lying position and holding
- Trigger control and firing a shot. Range Procedure and safety precautions. Aiming and alteration of sight. Theory of groups and snap shooting. Firing at moving targets. Miniature range firing. Characteristics of Carbine and LMG.
- Introduction to map, scales, and conventional signs. Topographical forms and technical terms.
- The grid system. Relief, contours, and gradients. Cardinal points and finding north. Types of bearings and use of service protractor. Prismatic compass and its use. Setting a map, finding north and own position. Map to ground and ground to map. Knots and lashings, Camouflage and concealment, Explosives and IEDs.
- Field defenses obstacles, mines and mine lying. Bridging, waterman ship. Field water supplies, tracks and their construction. Judging distance. Description of ground and indication of landmarks. Recognition and description of target. Observation and concealment. Field signals. Section formations. Fire control orders. Fire and movement. Movement with/without arms. Section battle drill. Types of communication, media, latest trends and developments OR

National Service Scheme (NSS-II) 1(0+1)

Evoking social consciousness among students through various activities viz., working together, constructive, and creative social work, to be skilful in executing democratic leadership, developing skill in program, to be able to seek self-employment, reducing gap between educated and uneducated, increasing awareness and desire to help sections of society. All the activities related to the National Service Scheme are distributed under four different courses viz., National Service Scheme I, National Service Scheme Il, National Service Scheme Ill and National Service Scheme IV each having one credit load. The entire four courses should be offered continuously for two years. A student enrolled in NSS course should put in at least 60 hours of social work in different activities in a semester other than five regular one-day camp in a year and one special camp for duration of 7 days at any semester break period in the two years. Different activities will include orientation lectures and practical works. Activities directed by the Central and State Government have to be performed by all the volunteers of NSS as per direction.

- Importance and role of youth leadership
- Meaning, types and traits of leadership, qualities of good leaders; importance and roles of youth leadership, Life competencies
- Definition and importance of life competencies, problem-solving and decision-making, interpersonal communication. Youth development programs
- Development of youth programs and policy at the national level, state level and voluntary sector; youth-focused and youth-led organizations
- Health, hygiene and sanitation. Definition needs and scope of health education; role of food, nutrition, safe drinking water, water borne diseases and sanitation (Swachh Bharat Abhiyan) for health; national health programs and reproductive health. Youth health, lifestyle, HIV AIDS and first aid. Healthy lifestyles, HIV AIDS, drugs and substance abuse, home nursing and first aid. Youth and yoga. History, philosophy, concept, myths, and misconceptions about yoga; yoga traditions and its impacts, yoga as a tool for healthy lifestyle, preventive and curative method.

Semester III

Crop Management in Natural Farming 3 (2+1)

Objectives

1. To acquaint students with the concepts of selection of crops and their cultivation practices under specific agroecological zones following natural farming principles
2. To train the students about package of practices of different crops under natural farming

Theory

Selection of crop and variety — Criteria for selection of crops in natural farming, Traditional and indigenous variety - Advantages and disadvantages - Classification of crops - factors affecting selection of crops and variety, Importance of pulses of legumes in natural farming, Weather management- Concept of weather and climate, Factors affecting crop production: climatic factors, edaphic, biotic, physiographic etc., micro-climate management techniques, Windbreaks and shelter belts, Management of weather in natural farms, Seed, sowing and Irrigation— Importance and concept of seed selection, seed treatment, Seed rate- Factors affecting seed rate and seed germination, sowing- time, method and depth, indigenous methods and animal drawn techniques, time and method of irrigation, management of irrigation, ITKs in seed, sowing and water management, Crop Geometry: Definition of crop geometry and crop density, types of crop geometry, importance of crop geometry, thinning, gap filling, optimum plant population, factor affecting plant population, relation of crop geometry and growth, relation of crop geometry and yield, Weed management: Classification of weeds and their importance, methods of weed management, practices of weed management in natural farming, Nutrient management: Sources of nutrient application and their management in natural farming and modern agriculture, maintenance of soil health and plant health, under natural farming, Package of practices of crops under natural farming.

Crop Management in Natural Farming 3 (2+1)

Practical

- Selection of crops for natural farming according to agro-ecological zones,
- Know how on grommet observatory and data interpretations,
- Calculation of seed rate for sowing,
- Plant geometry and population calculation,
- Traditional seed treatment methods,
- Preparation of traditional nutrient supply concoctions and their uses,
- Traditional knowledge techniques in crop management and on-farm waste management,
- Study of methods of crop residue management,
- Study of important methods of plant protection,
- Irrigation efficiency,
- Preparation of crop calendar for different sowing seasons under various micro-climates.
- Preparation of contingency crop plan for natural farming,
- Yield estimation in naturally grown crops, Field visit to natural farm,
- Documentation of sustainable natural farming practices.

Production of Bio-inputs2 (1 +1)

Objective

1. To impart knowledge on types of bio-inputs, their functions, compositions, application doses and impacts on soil processes
2. To develop skills among students in mass production of various bioinputs, their application methods and quality control

Theory

Definition and types of bioinputs, Importance of bioinputs, Mechanism of action of bioinputs, Quality standards of bioinputs, Methods of production and application doses of bioinputs, Factors affecting the product quality, Green manures, Achhadana, Cow-based bioproducts (Beejamrutha, Jeevamrutha, GhanJeevamrutha and other inputs (agniaster, brahmaster, neemaster, sonthaster, Khatti lassi, dasparni ark, Kanupajala and saptdhanyaankur) for pest, disease and nutrient management, compositions/ingredientsofvarious bioinputs and their relative merits, Effects of bioinputs on soil biological components, their activities, soil enzymes and overall soil fertility, Impact of bioinputs on plant physiological activities and physiological stress indicators, Influence of bioinputs on plantmicrobe and plant- faunal interactions, Initiatives taken by government(central/state), NGOs and other organizations in India for promotion of bioinputs under natural farming perspectives, Economic considerations, viability, marketing and export potential of different bioinputs, Quality aspects, grading, packaging, handling, and establishment of internal control system for bioinputs, Project formulation on bioinputs in natural farming.

Production of Bio-inputs2 (1 +1)

Practical

- Definition and types of bio inputs,
- Compositional analysis (nutritional and microbiological),
- Assay on mode of action of bio inputs,
- Preparation of animal manure-based bio inputs and their uses,
- Preparation of Beejamrutha, Jeevamrutha, Ghana Jeevamrutha, soil microbes based improved bio inputs,
- Mass production and its uses;
- Preparation of Bramhaster and their uses,
- Preparation of Agniaster, Neemaster and Dashparni ark, their mass production and uses,
- Preparation of plant-extract based bio inputs, Ginger-Garlic- Chilli extract and their use,
- Preparation of fermented butter milk and their uses,
- Preparation of herbal kunapajala and Neem -based concoctions and their specific uses,
- Study of physical, chemical and biological characteristics of important bio inputs,
- Determination of plant physiological stresses upon application of bio inputs.

Farm Power and Machinery 3 (1+2)

Objectives

I. To enable the students to understand the basic of farm power and machinery use in natural farming

2. To gain the knowledge and skills necessary to effectively and effciently manage farm machinery, optimize power utilization, and improve overall farm productivity.

Theory

Mechanization in agriculture: its potential and prospects, Sources of farm power, Draught animal power and its relevance to Indian Agriculture, Familiarization with different engine systems, Familiarization withdifferent systems ofI.C. engines, working principles of I C engines, Components of IC engine, Use of bio- diesel/bio-ethanol as alternative fuel for engines, Familiarization with tractor, Introduction to primary and secondary tillage equipment, Energy efficient improved tools and equipment's in natural farming: Introduction to rotavator, planter, vegetable transplanter, bed former, disk harrow, mulch laying equipment, Ergonomically improved hand tools for intercultural operations: wheel hoe, long handle hoe, garden tools, labour saving/drudgery reducing tools, Plant protection equipment: knapsack sprayer, power sprayer, threshing and shelling machinery, Animal drawn farm equipment/tools: Animal drawn bund former, Seed drill, Planters, cultivator, ITK in farm machinery, ITKS on farm tools.

Farm Power and Machinery 3 (1+2)

Practical

- Study of different engine systems, Study of the different components of IC engines,
- Familiarization with tractor, Familiarization with tools and equipment for input production in field,
- Familiarization with tractor drawn primary and secondary tillage implements,
- Familiarization with animal drawn primary and secondary tillage implements,
- Familiarization with tractor drawn seedcum- fertilizer drills their seed metering mechanism and calibration,
- Familiarization with animal drawn seed-cum-fertilizer drills their seed metering mechanism and calibration,
- Familiarization with planters,
- Familiarization with transplanters.
- Familiarization with different inter-cultivation equipment,
- Familiarization with different types of sprayers,
- Familiarization with different types of dusters,
- Familiarization with harvesting machinery,
- Familiarization with threshing machinery.

Livestock and Poultry Production 3 (2+1)

Objectives

1. To impart knowledge and skill in rearing oflivestock and poultry production
2. To make them aware about basics oflivestock and poultry production including managemental aspects under natural farming.

Theory

Historical account of livestock rearing in Vedic, medieval and modern era in India including free range, semi-intensive and intensive systems and major pastoral systems in India, Demographic distribution and characteristics of livestock (cattle, buffalo, sheep, goat, and poultry) and economy, Common farm management practices of livestock and poultry including housing, feeding, watering, sanitation, pest and disease management, Common vices of animals (cattle, buffalo, sheep, goat), their prevention and care. General principles of sewage disposal and purification, Solid and liquid manure management, Compost making. Ventilation requirements, Presumptive standards for permissible air pollution in buildings, Respiratory exchange in domestic animals, Importance and method of ventilation, Livestock and poultry rearing and carbon footprints, Breeding strategies and biotechnological tools for the improvement of dairy cattle, buffalo, sheep, goat, swine and poultry, Natural breeding for livestock and poultry, Methods of conservation — Livestock and poultry conservation program s in the state and the country, Fodder production and conservation Importance of grasslands and fodder in livestock production and their management, Structures for storage of feeds and fodders, Scarcity fodders and preservation of green fodder, Recycling of animal shed washings and wastes in fodder production and use of recycle waste., Disposal, utilization and recycling of waste, etc. Sheep, goat, rabbit farming and swine farming: role and importance in natural farming, Classification ofpoultry with respect to production characters, age and standards, Production characters of other avian species, Description of indigenous fowls and their value in rural farming, Livestock production management - Housing systems, layout and design of different buildings for animals, Selection of site: General principles affecting the design and construction of building for housing for various livestock species. Scavenging system of management — Low input technology, backyard and semi- intensive units; their management and economic achievements

Livestock and Poultry Production 3 (2+1)

Theory Cont------

Deep litter management — Control of litter-borne diseases and recycling of litter, Cage management — Different types; Advantages and disadvantages. Management of growers, layers, broilers and breeders, Stress management.

General management and feeding practices of calves, heifers, pregnant, lactating and dry animals, bulls and working animals. Systems of feeding, restricted feeding, phase feeding. Unconventional feed ingredients -Herbal Bio enhancers.

Protection and welfare of livestock, poultry and working animals. Draught ability of cattle and buffaloes; Routine animal farm operations and labour management, Animal farm accounts and records.

Indigenous technical knowledge related to livestock rearing and veterinary practices, Indigenous technical knowledge pertaining to health and management of livestock.

Classification of poultry with respect to production characters, age and standards; Description of indigenous fowls and their value in rural farming, Specific strains developed for rural poultry production their acceptability and importance in rural eco-system. Brooding management — Types ofbrooders, preparation of shed, Importance of environmental factors; Housing — Types of poultry houses, space requirements. Feeding management— Classification of nutrients, Nutrient requirements and feed formulations, Feeding systems; Water management.

Sheep and goat farming: Introduction and scope in the country, Important breeds (exotic and indigenous) and their characteristics, Housing and feeding. Management of sheep and goat for optimal production.

Principles of disease and pest management, Ayurvedic, homeopathic, ethno-medicines, General principles of medication for livestock and poultry in natural farming.

Candling, sexing, grading, packing and disposal of hatchery waste. Trouble shooting hatchery failures, Poultry waste management, pollution and environmental issues, mixed and integrated poultry farming under natural landscape. Vertical and horizontal integration in poultry production, Management of ducks, turkeys, Japanese quails and guinea fowls.

Livestock and Poultry Production 3 (2+1)

Practical

- Identification of indigenous breeds of livestock, small ruminants and poultry and other avians.
- Acquaintance with natural livestock farms (dairy and small ruminant), backyard poultry and other avians, Layout plans for different livestock houses and poultry, Fodder production and conservation: Visit to the fodder farm, Familiarization with various types of fodders in the states and India.
- Livestock waste utilization and recycling/ Familiarization with various fertilizers and manures. Human handling and restraining of cattle, buffalo, sheep and Goat: clipping, shearing, dipping, spraying and spotting sick animals,
- Determination of body weight using different measurements,
- Training Of breeding males: detection of heat, Identification and care of pregnant animals, Care of neonatal and young stock,
- Diagnosis, prevention and treatment of important diseases and pests of livestock, small ruminants and poultry and other avians, Economic traits of broilers, egg-type chicken and breeders,
- Poultry farm and hatchery equipment, Brooding arrangement in broiler farms,
- ITKS related to livestock management, Vaccination, deworming and medication of birds, Keeping records and their maintenance, Feed ingredients and its quality assessment, Fundamentals in poultry postmortem examination for sample collection and dispatch.
- Economics and preparation of project report for establishing livestock, fodder and poultry farms/festivals related to animals,
- Participation and documentation of the region-specific festivals related to animals.

Management of Plant Diseases2 (1 +1)

Objectives

1. To teach students the important plant diseases and plant pathogenic organisms.
2. To impart practical knowledge on crop loss assessment and disease management methods under natural farming.
3. To impart practical knowledge on the preparation of inputs for the management of plant diseases under natural farming.

Theory

Plant disease management under natural farming: Causes / factors affecting disease development, Disease triangle and tetrahedron and classification of plant diseases, Important plant pathogenic organisms, different groups: fungi, bacteria, fastidious vesicular bacteria, phytoplasmas, Spiro plasmas, viruses, viroids, algae, protozoa, phanerogamic parasites and nematodes , Pathogenesis, Koch's postulates, Role of enzymes, toxins and growth regulators in disease development, Defense mechanism in plants, Epidemiology: Factors affecting disease development, survey, surveillance and vigilance, Crop loss assessment and models, Principles and pre-requisites of forecasting of diseases, Principles and methods of plant disease management, Symptoms, etiology, disease cycle and management of major diseases of field crops and horticulture crops, Methods of plant disease control: Host plant resistance, cultural, mechanical, physical, legislative, biological control, Ecological management of crop environment, Role of soil microbiome in disease management, Suppressive soils, concepts and potentialities for managing soil borne pathogens, Application of khatti lassi, jungle ki Kandi, sontha aster for the management of various types of diseases, Plant growth promoting rhizobacteria (PGPR) and their use in plant protection, Heritage perspective of plant protection (Drumraksha), Plant Protection through Botanicals, ITKS related to management of plant diseases.

Management of Plant Diseases2 (1 +1)

Practical

- Acquaintance with various laboratory equipment and microscopy,
- Collection and preservation of disease specimens,
- Preparation of media, isolation and Koch's postulates,
- General study of different structures of fungi,
- Study of symptoms of various plant diseases,
- Study of representative fungal genera,
- Staining and identification of plant pathogenic bacteria,
- Transmission of plant viruses,
- Study of phanerogamic plant parasites,
- Preparation of extracts of natural plant products as fungicides,
- Field visit for the diagnosis of field problems,
- Collection and preservation of plant diseased specimens for Herbarium,
- Preparation of pressed and well mounted specimens,
- Preparation and methods of application of Khatti lassi, Jungle ki Kandi, Sontha aster for disease management, Biosurfactants and their use.

Biodiversity Conservation2 (1+1)

Objectives

I. To impart knowledge on biodiversity in different agro-ecological zones

2. To develop skills on assessment of biodiversity, procedures of biodiversity valuation and strategies of biodiversity conservation.

Theory

Definition, Genetic diversity, Species diversity, Ecosystem diversity: Structural and functional aspects. Agro ecological zones in India with a brief understanding of diversity in soil types, temperature, rainfall, watershed atlas of India, Seasons and season cycle in India, Festivals and seasons, Importance of local biodiversity, Bio-geographic classification of India, The value of biodiversity and conservation, India as a mega diversity nation, Hotspots of biodiversity: criteria for determining hot spots, Threats to biodiversity- habitat loss, pollution, species introduction, global climate change, overexploitation, poaching. Rare species, Extinction: mass extinction, extinction processes, Human factors: social, economic, political and remedial actions, Endangered and endemic species of India, common plant species, common animal species, Conservation of biodiversity- strategies for conservation: in-situ and ex-situ conservation, Conservation practices in India and world- organizations involved in resource conservation: IUCN, WWF, UNEP, UNESCO, Biodiversity International, IPGRI, FAO, BSI, ZSI. General account on activities of DBT, BSI, NBPGR, ZSI, FSI, NBFGR and NBAGR NFPTCR, Sacred groves, Biodiversity register, Conservation ofbiodiversity, Salient provisions of Biological Diversity Act, 2002, Role of Gramsabha in biodiversity conservation, Ancient Indian culture in biodiversity conservation, Role of biodiversity in SDGs, Ecological economics, Mission Life Style for Environment (Mission LiFE).

Biodiversity Conservation2 (1+1)

Practical

- Field survey for studying plant species diversity in a village/region,
- Study of the morphology and reproductive structures of the types of micro-flora and micro- fauna,
- Staining of fungal filaments by Cotton Blue, Methylene Blue,
- Preparation of sterilization media, principles of isolation, pure culturing, Collection, identification and submission of non-wood forest products,
- Visit to forest to study genus/species distribution,
- Notes on pest and diseases of forest plants (any four),
- Study of ITKs on biodiversity conservation and documentation of tangible and non-tangible benefit of biodiversity conservation,
- Orientation about biodiversity register and its activities.

Physical Education, First Aid, Yoga Practices and Meditation1 (0+1)

Objectives

1. To make the students aware about Physical Education, First Aid and Yoga Practices
2. To disseminate the knowledge and skill how to perform physical training, perform first aid and increase stamina and general wellbeing through yoga

Practical

Physical education; Training and Coaching - Meaning and Concept; Methods of Training; aerobic and aerobic exercises; Calisthenics, weight training, circuit training, interval training, Fartlek training; Effects of Exercise on Muscular, Respiratory, Circulatory &Digestive systems; Balanced Diet and Nutrition: Effects of Diet on Performance; Physiological changes due to ageing and role of regular exercise on ageing process; Personality, its dimensions and types; Role of sports in personality development; Motivation and Achievements in Sports; Learning and Theories of learning; Adolescent Problems and its Management; Posture; Postural Deformities; Exercises for good posture.

Yoga; History of Yoga, Types of Yoga, Introduction to Yoga,

- Asanas (Definition and Importance) Padmasan, Gaumukhasan, Bhadrasan, Vajrajasan, Shashankasan, Pashchimotasan, Ushtrasan, Tadasan, Padhastasan, Ardhchandrasan, Bhujangasan, Utanpadasan, Sarvangasan, Parvatasan, Patangasan, Shishupalanasan — left legright leg, Pavanmuktasan, Halasan, Sarpasan, Ardhdhanurasan, Sawasan
- Suryanamskar Pranayama (Definition and Importance) Omkar, Suryabhedan, Chandrabhedan, AnulomVilom, Shitali, Shitkari, Bhastrika, Bhramari
- Meditation (Definition and Importance), Yogic Kriyas (Kapalbhati), Tratak, Jalneti and Tribandh
- Mudras (Definition and Importance) Gyanmudra, Dhyanmudra, Vayumudra, Akashmudra, Pruthvimudra, Shunyamudra, Suryamudra, Varunmudra, Pranmudra, Apanmudra, Vyanmudra, Uddanmudra
- Role of yoga in sports *Teaching of Asanas — demonstration, practice, correction and practice.

Physical Education, First Aid, Yoga Practices and Meditation1 (0+1)

- History of sports and ancient games, Governance of sports in India; Important national sporting events; Awards in Sports; History, latest rules, measurements of playfield, specifications of equipment, skill, technique, style and coaching of major games (Cricket, football, table Tennis, Badminton, Volleyball, Basketball, Kabaddi and Kho-Kho) and Athletics.
- Need and requirement of first aid.
- First Aid equipment and upkeep.
- First AID Techniques, First aid related with respiratory system.
- First aid related with Heart, Blood and Circulation.
- First aid related with Wounds and Injuries.
- First aid related with Bones, Joints Muscle related injuries.
- First aid related with Nervous system and Unconsciousness.
- First aid related with Gastrointestinal Tract.
- First aid related with Skin, Burns.
- First aid related with Poisoning.
- First aid related with Bites and Stings.
- First aid related with Sense organs,
- Handling and transport of injured traumatized persons. Sports injuries and their treatments.

Skill Enhancement course 2 (0+2)

One Course of (0+2) credits of skill enhancement courses will be offered from indicated list of skill enhancement courses.

Environmental Studies and Disaster Management3 (2+1)

Objectives

1. To acquaint students with the significance of environmental principles and remediation procedures
2. To develop skills among students on environmental impact assessment

Theory

Multidisciplinary nature of environmental studies: Definition, scope and importance. Natural Resources: Renewable and non-renewable resources, Natural resources and associated problems, (a) **Forest resources:** Use and over- exploitation, deforestation, case studies, Timber extraction, mining, dams and their effects on forest and tribal people, (b) **Water resources:** Use and overutilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems, (c) **Mining resources:** Use and exploitation, environmental effects extraction and using mineral resources, case studies, (d) **Food resources:** World food problem, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies, (e) **Energy resources:** Growing energy needs, renewable and nonrenewable energy resources, use of alternate energy sources, case studies, (f) **Land resources:** Land as resources, land degradation, man induced landslides, soil erosion and desertification, Role of an individual in conservation of natural resources, Equitable use of resources for sustainable lifestyles. Ecosystem: Concept of ecosystem, structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem, Ecological succession, Food chains, food webs and ecological pyramids, Introduction, types, characteristics features, structure and function of the following ecosystems: (a) Forest ecosystem, (b) Grassland ecosystem, (c) Desert ecosystem d. Aquatic ecosystem (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity and its conservation: Introduction, definition genetic, species and ecosystem diversity and biogeographical classification of India. **Value of biodiversity:** consumptive use, productive use, social ethical, aesthetic and option values. Biodiversity at global, national and local level, India as a mega-diversity nation, Hot-sports of biodiversity, **Threats to biodiversity:** habitat loss, poaching of wildlife, man-wildlife, man-wildlife conflicts. Endangered and endemic spices of India.

Skill Enhancement course 2 (0+2)

Conservation of biodiversity: In—situ and Ex-situ conservation of biodiversity. Environmental pollution: Definition, cause, effects and control measures of: (a) Air Pollution, (b) Water Pollution, (c) Soil Pollution, (d) Marine Pollution, (e) Noise Pollution, (f) Thermal pollution e.g. nuclear hazards. Solid Waste Management: causes, effects, and control measures of urban and industrial wastes, Role of an individual in prevention of pollution.

Social issues and the environment: From Unsustainable to sustainable development, urban problem related to energy, water conservation, rain water harvesting, watershed management. Environmental ethics: Issues and possible solutions, climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust Dies, Wasteland reclamation, Consumerism and waste products, Environment Protection Act, Air (Prevention and Control of Pollution) Act, Water (Prevention and Control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation, public awareness.

Human Population and the Environment: Population growth, variation among nations, population explosion, Family welfare Program, Environment and human health: Human Rights, Value Education, HIV/AIDS, Women and Child welfare, Role of information technology in environment and human health.

Disaster Management: Natural Disaster: Meaning and nature of natural disasters, their types and effects. Floods, drought, cyclone, earthquakes, landslides, avalanches, volcanic eruptions, Heat and cold waves, climatic changes: global warming, sea level rise, ozone depletion. Man Made Disasters-Nuclear disasters, chemical disasters, biological disasters, building fire, coal fire, forest fire, oil fire, air pollution, water pollution, deforestation, industrial waste water pollution, road, rail, air and sea accidents.

Disaster Management: Effect to migrate natural disaster at national and global levels, International strategy for disaster reduction, Concept of disaster management, national disaster management framework, financial arrangement, role of NGOs, community-based organization and media, Central, state, district and local administration, Armed forces in disaster response, Disaster response, Police and other organizations.

Semester IV

Elements, Characteristics and Design of Natural Farming Systems 3 (2+1)

Objectives

I. To make students aware about concept, function and design of natural farming as a system approach

2. To train students on different advanced methodologies for analyzing and designing agroecological specific natural farming units.

Theory

Natural Farming Systems: Concept, importance, approaches, determinants and characteristics, Strength and weakness of natural farming systems, Ecosystems: Concept and components, System approach and concept for farming systems, natural farming systems and ecosystem analysis, The water cycle, nutrient Cycle and energy cycle, Types of ecosystems: Natural ecosystems and agriecosystems, concept and classification of ecosystem services, Concept and elements of natural systems, Classification of farming types (based on geographical condition, levels of technology and external input use, labour, demand of produce, etc.), Subsistence farming, Intensive subsistence and primitive subsistence, Traditional farming, intensive conventional farming, integrated farming, conservation agriculture, eco-agriculture, ecologically intensive agriculture (Sustainable intensifications) biodynamic farming, permaculture, cow based farming, cosmic energy farming, prakritik shaswat yogic farming, natural farming, agnihoter farming (homa farming), astrology and astronomy in agriculture, vastu practices, Important agri-based, horti-based, animal based, agro forestry, aqua farming system models, The prairie model, Natural system models at the basis of mimicry (the rain forest model, the dry forest model, Concept of ecosystem based integration of crops, livestock, horticulture and fish farming as complimentary activities, Traditional integrated farming system models based on natural farming principles, Comparison of natural ecosystems, traditional farming system and modern agriculture, Production syndrome in natural farming, Hypotheses and concepts in designing the natural farming system models (Biodiversity and the mimicry hypothesis, productivity, resilience, equilibrium and stability, Levels of organization for nature mimicry in agro-ecosystems), Principles for cropping and farming system design based on natural ecosystem mimicry, Steps for development of agricultural systems based on mimicking natural ecosystems, Economic evaluation and comparative analysis of different farming systems and natural systems, optimization methodology: a potential approach for system, natural farming system and ecosystem analysis and design.

Elements, Characteristics and Design of Natural Farming Systems 3 (2+1)

Practical

- Identification and characterization of different natural and agro-ecosystems in the rainfed, hill and mountain, arid, coastal and irrigated regions,
- Study on characterization of natural eco-systems and intensive farming system,
- Study on characteristics of elements of water cycle in natural ecosystem and farming system,
- Study on characteristics of elements of nutrient cycle in natural ecosystem and farming system,
- Study on characteristics of elements of energy cycle in natural farming,
- Comparative biodiversity analysis in natural and conventional farming systems,
- Calculation of biodiversity index for different farming systems,
- Designing of cropping/farming and natural farming systems,
- Study and representation of resource flow and recycling in different farming systems,
- Budgeting: Farm records and farm book keeping in natural farming system,
- Cost and profit analysis of different components and whole system,
- Designing of natural farming systems models in different agro-ecosystems,
- Designing farming systems/natural farming systems using mathematical tools/optimization methodology.

Improvement of Traditional and Underutilized Crops 3 (2+1)

Objectives

1. To gain the knowledge on different types of traditional and underutilized crops
2. To train the students on the methods of improvement techniques.

Theory

Importance of traditional and underutilized crops, Genetic basis of crop Improvement: Laws of inheritance; qualitative and quantitative traits, multiple factor hypothesis, pureline theory, components of phenotypic variance, GXE interaction, heritability, breeder's equation, populations/ landraces/ farmers' varieties/ cultivars, Hardy Weinberg equilibrium and factors affecting it, Assortative and disassortative mating, Difference between self-pollinated, cross pollinated and clonal populations, Important physiological parameters for crop improvement- Photosynthesis, transpiration, water and mineral Details of the Course regulation, Breeding objectives under natural farming systems- Nutrient Use Efficiency, Tolerance to micronutrients, mineral deficiencies and toxicities, Weed competitiveness, Enhanced Interaction with Rhizospheric microorganisms, Multiple disease and insect resistance, Quality, Earliness/ crop duration to suit multiple cropping system and agro- ecologies, Wider adaptability and stability, Broad genetic base / population heterogeneity, High yield, Multipurpose crops- stay green, food, feed, fiber and fuel. Plant breeding techniques: Domestication, introduction, Selection, hybridization; Breeding self-pollinated cropsPureline selection, mass selection, pedigree selection, mass selection, SSD, backcross breeding, Breeding cross pollinated crops- Population improvement, recurrent selection, composite and synthetic varieties, Improving clonally propagated crops- Clonal selection, Farmer participatory plant breeding. Molecular markers, Marker Assisted Selection (MAS), comparison of MAS vs. Phenotypic selection.

Improvement of Traditional and Underutilized Crops 3 (2+1)

Practical

- Study of floral biology of different crops,
- Hybridization techniques, DUS characterization important land races and traditional varieties of crops,
- Breeding methods for natural farming,
- Study of root characteristics of crops grown under natural and other production systems,
- Learning of techniques of participatory plant breeding,
- Techniques of conservation of plant genetic resources,
- PRA for mapping indigenous crops and their characteristics,
- Seed identification and collection for natural farming systems,
- Identification of important land races and traditional varieties of crops,
- Identification of important land races and traditional varieties of fruit crops.

Agro-Eco System Analysis for Natural Farming 2 (1 +1)

Objectives

I. To enhance the knowledge about the concepts, types and tools adopted in agro-ecosystem

2. To study the techniques and strategies adopted in natural farming and as a tool in agroecosystem analysis

Theory

Definition of system, system hierarchy, Introduction, and concept of agro- ecosystem, Properties of agro-ecosystem, Types of agro ecosystem analysis, Tools and techniques used in agroecosystem analysis, agro-ecosystem analysis and sustainable agriculture, Steps involved in agroecosystem analysis, Methodology of agro-ecosystem analysis, Energy flow of agro-eco system analysis, Participatory rural appraisal related terms — RRA, PRA, PLA, PLAM, and philosophy of PRM, Participatory planning framework, Importance of participation, Principles of participatory extension, Participatory technology development process

Practical

- Simulated exercise on space related methods,
- Time related methods, Flow related methods,
- Documentation of participatory technology development process, conducting agro-ecosystem analysis using PRA tools and techniques in village situation,
- Learn the techniques of action plan development.

Beneficial Insects 3 (2+1)

Objectives

1. To teach students the role of beneficial insects viz., pollinators, decomposers, predators and parasitoids, soil builders, productive insects
2. To impart knowledge on the beneficial insects used under natural farming

Theory

Ecological understanding of insects in natural farming, Study of their life cycle, population dynamics and interaction with other abiotic and biotic components in a natural farming ecosystem, Decomposers (insects) in natural farming system, Pollinators in natural farming systems: their kinds, types, diversity, conservation and profitable use for enhancing.

Productivity in terms of quality and quantity, Natural enemies of poriferous insect's generalist predators, specific predators, parasitoids (egg, larval, pupal and intermediary), Insects in ecological communities, Insects as food, soil builders and insects of aesthetic value, Promotion of required habitat for friendly insects, Pollinators: Honey -bee biology, commercial methods of rearing, equipment used, seasonal management, bee enemies and diseases, Bee pasturage, bee foraging and communication, Insect pests and diseases of honey bee, Role of pollinators in cross pollinated plants, Types of silkworms, voltinism and biology of silkworm, Mulberry cultivation, mulberry varieties and methods of harvesting and preservation of leaves, Rearing, mounting and harvesting of cocoons, Pests and diseases of silkworm, management, rearing appliances of mulberry silkworm and methods of disinfection, Species of lac insect, morphology, biology, host plant, lac production — seed lac, button lac, shellac, lac- products, Indigenous technical knowledge on management of insect pests, Knowledge and importance of beneficial insects in ancient India.

Beneficial Insects 3 (2+1)

Practical

- Rearing dung beetles for decomposing organic wastes as manure,
- **Pollinators:** Diversity, species richness and their use in enhancing productivity of crops,
- **Natural enemies- diversity:** species richness, their conservation in in-situ and wherever possible, artificial rearing for using pest management.
- Collection of beneficial insects,
- Study of life cycle and habitats of beneficial insects,
- Crop wise applications of beneficial insects for pest control.

Quantification and Valuation of Ecosystem Service 3 (2+1)

Objectives

I. To orient the students with basic concepts of ecosystems services

II. To impart skills on quantification and valuation of ecosystem services

Theory

Ecosystem Services (ES)-Meaning, Concept and its importance, Classification of Ecosystem Services - Provisioning, Regulating, Supporting and Cultural services, Basics of natural capital, Quantification of ecosystem services-Direct and Indirect approaches, Valuation of Ecosystem Services and its need, Ecosystem valuation methods-Revealed preference methods: Market pricing, Production function,Hedonicpricingmethods, Travel cost method and Random utility models, Stated preference methods-Contingent valuation method and choice modelling, Cost based approaches of Ecosystem Valuation- opportunity cost, Cost of alternatives or substitute goods, Replacement cost method, Methods for obtaining Non- economic Values-Focus groups, Citizens' Juries, Health based valuation , Q-methodology and Delphi surveys, Payment for ecosystem services (PES), Governance and policy issues in ecosystem services, Challenges in valuation of ecosystem services.

Quantification and Valuation of Ecosystem Service 3 (2+1)

Practical

- Ecosystem Valuation methods- Direct and indirect methods of calculation,
- An overview Millennium Ecosystem Services (MEA) Assessment,
- Case studies on payment for ecosystem services (PES),
- Case studies in Ecosystem Services in India and abroad,
- Study on Environmental Impact Analysis.
- Visits to the ecosystem areas (Agro ecosystem, Forest Ecosystem and Aquatic ecosystem).

Agricultural Marketing and Trade 3 (2+1)

Objectives

l. To understand the fundamentals of agricultural marketing and trade 2. To analyze the factors influencing supply and demand in agricultural markets

3. TO explore different marketing channels and strategies in agriculture
4. To examine the role of government policies and regulations in agricultural markets

Theory

Agricultural Marketing: Concepts and definitions of market, marketing, agricultural marketing, market structure, marketing mix and market segmentation, classification and characteristics of agricultural markets; demand, supply and producer's surplus of agri commodities: nature and determinants of demand and supply of farm products, producer's surplus — meaning and its types, marketable and marketed surplus, factors affecting marketable surplus of agricommodities; pricing and promotion strategies: pricing considerations and approaches — cost based and competition based pricing; market promotion — advertising, personal selling, sales promotion and publicity — meaning, merits and demerits; marketing process and functions: Marketing process concentration, dispersion and equalization; exchange functions — buying and selling; physical functions — storage, transport and processing; facilitating functions — packaging, branding, grading, quality control and labelling (Agmark); Market functionaries and marketing channels: Types and importance of agencies involved in agricultural marketing; meaning and definition of marketing channel; number of channel levels; marketing channels for different farm products; Integration, efficiency, costs and price spread: Meaning, definition and types of market integration; marketing efficiency; marketing costs, margins and price spread; factors affecting cost of marketing; reasons for higher marketing costs of farm commodities; ways of reducing marketing costs; Role of Govt. in agricultural marketing: Public sector institutions- CWC, SWC, FCI, CACP and DMI — their objectives and functions; cooperative marketing in India:

Agricultural Marketing and Trade 3 (2+1)

Theory Cont---

Risk in marketing: Types of risk in marketing; speculation and hedging; an overview of futures trading; Agricultural prices and policy: Meaning and functions of price; administered prices; need for innovations in agricultural price policy; Trade: Concept of International Trade and its need, theories of absolute and comparative advantage. Present status and prospects of international trade in agri-commodities; WTO; Agreement on Agriculture (AoA) and its implications on Indian agriculture; IPR. Role of government in agricultural marketing. Role of APMC and its relevance in the present day context.

Practical

- Plotting and study of demand and supply curves and calculation of elasticities;
- Study of relationship between market arrivals and prices of some selected commodities;
- Computation of marketable and marketed surplus of important commodities;
- Study of price behaviour over time for some selected commodities;
- Construction of index numbers;
- Visit to a local market to study various marketing functions performed by different agencies, identification of marketing channels for selected commodity, collection of data regarding marketing costs, margins and price spread and presentation of report in the class;
- Visit to market institutions —NAFED, SWC, CWC, cooperative marketing society, etc. to study their organization and functioning.
- Application of principles of comparative advantage of international trade.

Agricultural Informatics and Artificial Intelligence 3 (2+1)

Objective

1. To acquaint students with the basics of computer applications in agriculture, multimedia, database management, application of mobile app and decision- making processes, etc.
2. To provide basic knowledge of computer with applications in Agriculture
3. To make the students familiar with Agricultural-Informatics, its components and applications in agriculture

Theory

Introduction to Computers, Anatomy of Computers, Memory Concepts, Units of Memory, Operating System: Definition and types, Applications of MS-Office for creating, Editing and Formatting a document, Data presentation, Tabulation and graph creation, Statistical analysis, Mathematical expressions, Database, concepts and types, creating database, Uses of DBMS in Agriculture, Internet and World Wide Web (WWW): Concepts and components.

Computer programming: General concepts, Introduction to Visual Basic, Java, Fortran, C/ C++, etc. concepts and standard input/output operations.

e-Agriculture, Concepts, design and development, Application of innovative ways to use information and communication technologies (IT) in Agriculture, Computer Models in Agriculture: Statistical, weather analysis and crop simulation models, concepts, structure, inputsoutputs files, limitation, advantages and application of models for understanding plant processes, sensitivity, verification, calibration and validation, IT applications for computation of water and nutrient requirement of crops, Computer-controlled devices (automated systems) for Agri-input management, Smartphone mobile apps in agriculture for farm advice: Market price, postharvest management etc., Geospatial technology: Concepts, techniques, components and uses for generating valuable agri-information, Decision support systems: Concepts, components and applications in Agriculture, Agriculture Expert System, Soil Information Systems etc. for supporting farm decisions. Preparation of contingent crop-planning and crop calendars using IT tools, Digital India and schemes to promote digitalization of agriculture in India.

Agricultural Informatics and Artificial Intelligence 3 (2+1)

Practical

- Study of computer components, accessories, practice of important DOS Commands,
- Introduction of different operating systems such as Windows, Unix/ Linux, creating files and folders, File Management. Use of MS-WORD and MS Power-point for creating, editing and presenting a scientific document,
- MS- EXCEL - Creating a spreadsheet, Use of statistical tools, Writing expressions, Creating graphs, Analysis of scientific data, Handling macros.
- MS-ACCESS: Creating Database, preparing queries and reports, Demonstration of Agri- information system,
- Introduction to World Wide Web (WWW) and its components, Introduction of programming languages such as Visual Basic, Java, Fortran, C, C++, Hands on practice on Crop Simulation Models (CSM), DSSAT/Crop-Info/Crop Syst/ Wofost,
- Preparation of inputs file for CSM and study of model outputs, computation of water and nutrient requirements of crop using CSM and IT tools,
- Use of smart phones and other devices in agro-advisory and dissemination of market information,
- Introduction of Geospatial Technology, Hands on practice on preparation of Decision Support System, Preparation of contingent crop planning,
- India Digital Ecosystem of Agriculture (IDEA).

SEC-VI

Skill Enhancement Course2 (0+2)

One Course of (0+2) credits of skill enhancement courses will be offered from indicated list of skill enhancement courses.

Semester V

Post Harvest Management 3 (2+1)

Objectives

1. To develop skills for primary and secondary processing, mechanical and biocontrol practices for storage of fruits and vegetables
2. To create awareness about natural preservation methods and hygiene and handling of fruits and vegetables.

Theory

Post-harvest management: Importance of fruits and vegetables, Introduction to post- harvest physiology of fruits and vegetables; Post-harvest losses, Maturity indices, Harvesting methods and collection devices, Unit operations in packaging with emphasis on use of natural products like bee wax, gum Arabic, shellac, xantham gum etc., Environment friendly and safe ripening methods, Post-harvest diseases, Disorders and their management.

Storage techniques: Traditional and modern storage structures: on farm and off farm storage, refrigerated storage, evaporative cool chambers, refrigerated vehicles, Storage techniques for local cultivars seeds.

Value addition: Principles and methods of preservation, natural preservatives, indigenous and traditional preservation techniques, phytochemicals for preservation, FSSAI, BIS, and Codex standards for major processed products from fruits and vegetables Value addition in fruits and vegetables through drying, pickling, jam, jelly, marmalade, preserve (murabba), candy, juice, ready-to-serve, squash, nectar, crystallized products, etc.; minimal processing, Eco-friendly handling and packaging: Use of essentials oils and ecofriendly phytochemicals in postharvest handling; Edible films and coating, eco or bio- based polymeric films used for packaging.

Modified atmosphere packaging, Vacuum packaging, Smart packaging, Active packaging using natural or bio-colours, ITKs for handling and packaging; neem- based products for fruit and vegetable handling.

Post Harvest Management 3 (2+1)

Practical

- Demonstration and use of different harvesting tools,
- Practical demonstration on packhouse operations,
- Study of different preservation methods for fruits, and vegetables,
- Study of weight loss of produce in eco-friendly packaging,
- Determination of TSS and acidity in fresh and processed products,
- Determination of ascorbic acid, Determination of sugars,
- Study of different types of dryers, Study of evaporative cooling system/low cost cooling system/ Zero Energy Cool Chamber,
- Preparation of traditional value-added products from fruits and vegetables,
- Preparation of jam, Preparation of jelly,
- Preparation of fruit-based beverages,
- Preparation of pickles,
- Study of different packaging materials,
- Visit to industry/processing unit.

Bio-resources and Agricultural Waste Management 3 (2+1)

Objectives

I. To impart knowledge on various aspects of biowastes and agricultural wastes for managing as organic amendments in agricultural production system

2. To develop skills on various methods of conversion of decomposable wastes into organic manures

Theory

Introduction to bio-resources and agricultural biomass waste, Classification and characterization of agri and livestock biomass waste, Principles of agricultural waste management: 4R's approach, Potential of recyclable crop residues and its management, In-situ management of agricultural waste, Role of soil and plants in waste management, Impact of agro-waste on soil and plant quality and the environment, Pre-treatment of agricultural wastes, Pre-treatment methods: biological treatment, biological-processes of waste management, Bio- conversion/ bio-decomposition through native microflora biogas generation, operation and management of biogas plants, utilization of biogas and spent slurry, landfill, Farm waste management machinery, Environmental benefit of waste management, ITK information on waste management. Mission Life Style for Environment (LiFE) and Govt. schemes to promote agriculture waste into wealth.

Practical

- Visit to various agri-farms,
- Collection and preparation of agricultural waste samples,
- Characterization of agricultural waste,
- Determination of ash content of agricultural wastes and determination of un-burnt carbon in ash,
- Determination of pH, EC and CEC, Determination of BOD and COD, Nutrient status (N, P, K, and micronutrients) analysis of agricultural waste,
- Survey of different agri -waste from livestock, dairy, food processing, fruit and vegetable and agri-chemicals,
- Study of biogas production process and Study briquetting of agricultural residues and important commercial agri-waste management methods in public and private sector.

Medicinal and Aromatic Plants 2 (1+1)

Objectives

I. To impart knowledge on diversity of medicinal and aromatic plants and their utilities

2. To develop skilled manpower on cultivation practices of medicinal and aromatic plants under natural farming system

Theory

General aspecs of Medicinal Plants: Definition, history, present and future needs, Introduction of plant parts (fruit, leaves, roots, stem, seeds and their modification), Cultivation and harvesting practices, Processing and storage practices, Marketing of medicinal products, Important Indian medicinal plants, Plant parts used as powder: Identification and utilization of Amla (Embelica offcinalis), Behra (Terminalia bellerica), Harad (Terminalia chebula), Turmeric (Curcuma longa), Garlic (Allium sativum), Bitter guard (Momordica charantia), Black plum (Syzygium cumini), Fenugreek (Trigonella foenumgraecum), Cinnamon (Cinnamomum verum), Sarpgandha (Rauvolfia serpentina), Black pepper (Piper nigrum), Ashwagandha (Withania somnifera), Psyllium husk (Plantego ovata). Plant parts used as juice/ decoction: Identification and utilization of Amla (Embelica offcinalis), Ginger (Gingiber officinalis), onion (Alium cepa), Bottle guard (Lagenaria siceraria), Basil (Oscimum santum), Arjun (Terminalia arjuna Neem (Azadirecta indica), Gwarpatha (Aloe vera), Brahmi (Bacopa monnieri), Giloy (Tinospora cordifolia), Shankhpushpi (Convolvulus prostrate), Bael (Aegle marmelos). Plant parts used as Lotion and Ointments: Identification and utilization of Gwarpatha (Aloe vera), Fenugreek (Trigonellafoenumgraecum), Pot marigold (Calendula officinalis), Neem (Azadirecta indica) Plant Parts Used as Oil: Clove (Syzygium aromaticum), Neem (Azadirecta indica), Coconut (Coccus nucifera), Nilgiri (Eucalyptus sp.). Plant parts used as Surgical Fibres, Sutures and Dressings: Identification and utilization of cotton (Gossipium sp.), Jute (Corchorus capsularis), Banana (Musa sp.). Plant parts used as Poultice: Identification and utilization ofTurmeric (Curcuma longa), Nilgiri (Eucalyptus sp.), Ginger (Gingiber officinalis), Garlic (Allium sativum), onion (Alium cepa), Dhatura (Dhatura sp.), Aak (Calotropis sp.), Arandi (Ricinus communis).

Practical

- Identification of locally available common medicinal plants,
- Basic preparation of herbal products as kadha, powder (Ex. Neem leaf, moringa leaf, tulsi leaf, giloy, arandana), Juice (Ex. Amla, Aloe vera), Trifala, Chyavanprash, Amla candy, herbal tea, etc.
- Study and documentation of commercial production of at least five medicinal plants, (Using website/ you Tube).
- Submission of digital photo album of at least ten medicinal plants with brief description,
- Cultivation maintenance and reporting of at least five medicinal plants within college campus.

Seed Production Technology 3 (2+1)

Objectives

1. To make the students aware about concept of seed and quality of seed
2. Principles of seed production of important cereals, pulses, oilseeds, fodder crops vegetables
3. To make aware about seed legislation, seed certification standards and seed processing.

Theory

Seed quality: Definition and concept of seed quality, Classification of seeds, Use of biodiversity for seed production, Formal and informal seed production systems. Principles of seed production of important cereals, pulses, oilseeds, fodder crops vegetables, Seed Physiology: Seed germination, viability, seed dormancy, types and requirements of seed germination, Seed health, quality seed collection from natural farming, Seed legislation, Seed testing, Varietal identification through Grow Out Test and electrophoresis, Biochemical tests and molecular tools, Minimum Seed Certification standards, field standards and seed standards, Seed drying and processing, Orthodox seed and recalcitrant seeds, General principles, stages and factors affecting seed longevity during storage and handling, Seed Village Concepts (seed self reliance), types and significance of indigenous seed conservation, community seed banks, Seed marketing, Role of WTO and oECD in seed marketing, Seed production, collection, storage and quality control in context with Natural farming, Technology of Seed Health during ancient time, Materials recommended (plant based, animal based) in ancient time for seed treatments, Sample case study of a few farmers initiative on conservation of seed in different states of India, PPVFR Act (2001) and Suo generis plant variety law approach, ITKs in seed storage and seed health as compiled by ICAR.

Seed Production Technology 3 (2+1)

Practical

- Seed production in major self-pollinated crops,
- Seed production in major cross pollinated crops,
- Seed production in selected vegetable crops,
- Seed sampling procedures,
- Seed testing: Physical purity, moisture test, varietal purity.
- RPL, germination, viability,
- Grow out test and electrophoresis, etc.
- Seedling vigour tests, Seed health tests,
- Genetic purity test: moisture tests, heterogeneity,
- Seed certification: Procedure, field inspection, taking of field counts,
- Preparation of field inspection and seed testing reports,
- Seed collection and rural improved storages,
- Visits to seed production farms, seed testing laboratories and seed processing plants.

Value Chain Management in Natural Farming 2 (1+1)

Objectives

1. Role and importance of value chains, value chain models
2. Role of Farmer Producer organization in value chains of natural farming

Theory

Concept of agriculture value chain, characteristics of value chain management in natural farming, types of value chains, role and importance ofvalue chains; dynamics of renewable and non renewable resources, resource scarcity, pricing and valuation of natural resources; good agriculture practices, good manufacturing and good processing practices, Value chain models- Producer driven, buyer driven, facilitated models.

Role of farmer producer organizations in value chains of natural farming products; Market initiation and strategies, organization of value chains, roles and responsibilities of stakeholders of value chain, transportation, logistics and infrastructure, cold chain components, dispute resolutions and arbitration, physical asset collateralization, Quality certification like HACCP and FSSAI standards, Postharvest, quality, and value-added aspects for domestic and export market special economic zones, Theory of storage and Warehouses — Types, classification, advantages and disadvantages.

Practical

- Case studies and success stories on natural farming or chemical free farming value chains in India and abroad,
- Environmental resource accounting techniques,
- Visit to special economic zones,
- Identification of value chains for commodities,
- Identification of niche markets for natural farming produces;
- Post-harvest loss assessment.

Renewable Energy Sources 2 (1+1)

Objectives

1. To gain the knowledge on different types of renewable energy sources
2. To understand the importance of renewable energy technology and its applications

Theory

Introduction to energy, Forms of energy, Conservation of energy, Sources of energy and their classification, Energy consumption patterns in India, Concept of renewable energy sources, Potential of renewable energy sources, Classification of renewable energy sources, Ecological footprint, Carbon footprint. Role of renewable energy in natural farming and its impact on food system, Environment and economy, Solar Energy: Introduction of solar energy, solar thermal energy systems: solar cooker, solar distillation, solar water heater, solar dryer, solar photovoltaic systems and their different types and uses, soil solarization, Biomass Energy: Introduction of biomass, biogas—biogas generation process, types of biogas plants, application of biogas, usage of biogas spent slurry, Biomass Gasification- types of gasifiers, producer gas production and utilization, Introduction to ethanol, biodiesel and hydrogen production process. Wind Energy: Introduction, Traditional practices of wind energy, Wind mills and their applications, Role of renewable energy in carbon mitigation, Energy balance in natural farming, Programs and policies for promotion of renewable energy in India.

Renewable Energy Sources 2 (1+1)

Practical

- Visit to farms to study different source of energy and their consumption,
- Study of solar cooker, Study of solar dryer,
- Study of solar water heater, Study of solar distillation,
- Study of solar photovoltaic system, Study of biogas plant,
- Study of biomass gasifier,
- Study the production process of biodiesel,
- Study of improved biomass cookstove,
- Traditional practices of wind energy,
- Determination of energy balance in natural farming.

Natural Farming and Human Health

Objectives

1. To make the students aware about concept of healthy food, Nutritional garden, Homestead kitchen garden and Nutri smart village
2. To understand the beneficial effects of medicinal crops on health

Theory

Changing lifestyles and food habits of the people over the years, Health and disease statistics of world and India, Food production and requirement vis-å-vis growth in human population in India and World, Need of healthy food, Concept of healthy food, safe food, organic food, green food, pesticide free food, fast food, slow food, Dietary guidelines, Traditional food visa-vis modern food and its potential health impact, Food Chain contamination and human health hazards, Deficiency in micro-nutrients by the human and livestock population, Malnutrition, Concept of one health, Traditional food of Indian population: Food and its nutrient content, Traditional diets in Indian and their adequacy- Nutritional value of traditional food, Concept of nutrition and health per acre, Interrelationship between natural farming and human health, Health and nutrition disconnect in the agriculture policy of India, Effect of natural farming on animal reproduction, Nutrients, taste, safety, immune system, antibiotic resistance, antioxidant value and protection of environment, Dietary requirement for Indian population: Current diet and nutrition scenario, Recommended Dietary Allowances (RDA): Food exchange list, food composition database, food composition and food groups, bio- fortification of crops, Criteria for classification of quality and safe food, Human health — Definition and philosophy, concept of health- biomedical, ecological, psychological and holistic, Criteria and qualities of organic and natural food, Comparison of natural and conventionally grown foods: Nutritional and nonnutritional components, bioactive components, their impact on human health, effect of processing on nutritional components, conservation of nutrients in processed foods, Food standards, food laws and labelling of natural foods, Concept of nutritional garden, homestead kitchen garden and nutri smart village. ITKS in human food and nutrition and health, Mission Life Style for Environment (LiFE).

Natural Farming and Human Health

Practical

- Physical examination of food, determination of gluten, bulk density, hydration capacity and index, oil absorption capacity, Sensory evaluation and organoleptic evaluation of natural and conventional foods,
- Analysis of proximate composition in conventional foods and natural farming produce/foods based on available secondary data.
- Determination of pesticide residues and heavy metal contamination in foods,
- Market survey and listing of natural farming foods available in the market,
- Preparation of nutri- thali,
- Composition of nutrient in different crop plants,
- Effect of processing on nutritive value of food,
- Determination of pesticides load in the food stuffs,
- Glycemic index of common food, Macro — nutrient and micro — nutrient and their sources,
- Calorie requirement among different class of population,
- Balance diet for different class of people,
- Drinking water standard - Limits of heavy metal content

Fundamentals of Entomology 3 (2+1)

Objectives

1. To teach the students on external morphology and anatomy of different systems of insects
2. To teach the students on classification and identification of insects upto family level

Theory

Entomology in ancient India, Historical classics of modern entomology, Classification of phylum Arthropoda up to classes, Relationship of class Insecta with other classes of Arthropoda, Major points related to dominance of Insecta in animal kingdom. Morphology: Structure and functions of insect cuticle and moulting, Body segmentation; Structure ofhead, thorax and abdomen. Structure and modifications of insect antennae, mouth parts, legs, wing venation, modifications and wing coupling apparatus, Structure and modifications of abdominal appendages, Anatomy of insect systems - Digestive, excretory, respiratory, circulatory, nervous, reproductive and glandular systems, sense organs and tropism and bio-communication in insects, Metamorphosis in insects, Taxonomy — Importance, history and development and binomial nomenclature, Definitions of Bio-types, Subspecies, Species, Genus, Family and Order, Classification of class Insecta up to Orders, basic groups of present days insects with special emphasis to orders and families of economic importance and their distinguishing characters.

Fundamentals of Entomology 3 (2+1)

Practical

- External features of Grasshopper/Blister beetle,
- Methods of collection and preservation of insects including immature stages,
- Types of insect antennae, mouthparts, legs, wings, wing venation and wing coupling apparatus,
- Types of insect larvae and pupae,
- Dissection of digestive system in insects,
- Dissection of male and female reproductive systems in insects,
- Study on characters of orders orthoptera, Dictyoptera, odonata, Isoptera,
- Thysanoptera, Hemiptera, Lepidoptera, Coleoptera, Diptera, Hymenoptera, Neuroptera, and their families of agricultural importance,
- Identification of major pests of crops.

Semester VI
Indian Traditional Knowledge 3 (2+1)

Objectives

1. To equip the students with importance of Indian Traditional Knowledge
2. To develop skill for integration of traditional practices for the benefit of farming community

Theory

Introduction and importance of Indian Traditional Knowledge in agriculture and allied sciences: Definition, Difference between Traditional Knowledge System and Western Science System, Protection of Traditional Knowledge, IPR and other provisions, Indian Traditional Knowledge in Agriculture covering Veterinary and Animal Husbandry, Pest and Disease Management, Grain/ Seed storage, Horticultural Crops, Crops and Cropping systems, Farm implements, Weather forecasting, Soil and water conservation, Soil fertility management, Rain water management, Tillage practices, Fisheries, Post-Harvest Technology, Garbage disposal and management, Wind erosion, Waste water management.

Indian Traditional Knowledge3 (2+1)

Practical

- Case studies on validation of ITKs,
- Need to revive traditional technologies relevant to the contemporary agricultural scenario,
- Geographical indications of plant species involved in various ITK's (Pest and disease management,
- Grain/ Seed Storage, Veterinary and Animal Husbandry and Weather forecasting).

Post-Harvest Management -II 3 (2+1)

Objectives

1. To develop skills for primary and secondary processing, mechanical and biocontrol practices for storage pest management in cereals, pulses and oil seeds
2. To create awareness about natural preservation methods and hygiene and handling Of fish and meat

Theory

Physico-chemical and nutritive properties of cereals, pulses and oilseeds, causes of postharvest losses, traditional and modern storage structures, on farm and off farm storage, bulk and bag storage (especially eco/bio-based bags), Primary processing of cereals, pulses and oilseeds: Cleaning, grading, curing/tempering/conditioning, pretreatments including parboiling, Secondary processing: Drying, size reduction (milling), Oil seed milling: Ghanis, hydraulic presses, expellers, extrusion processing and different types of extruded products (snack, breakfast cereals and weaning foods), bakery and confectionary products, Traditional value-added products (regional products like nuggets/wadi/badi, papad, vermicelli/sevai, etc.), Storage Pest Management: Use of ITKs like oil treatment, ash treatment, bio- control practices for storage pest control of cereals, pulses and oilseeds, Primary processing operations in poultry: Pre-slaughter operations and slaughtering operations for poultry: stunning, icing, grading, blanching, washing, evisceration, beheading, scaling, cutting off fins and belly flaps, steaking, filleting, skinning, deboning (meat-bone separator), mincing of skinned fillets, peeling, deveining, shucking, knobbing, Preservation methods for fish: Preservation of meat and fish by canning, chilling, freezing, marination (pickling), curing, cooking and smoking, dehydration, and biological preservatives (fermentation), etc. Value-added products from fish and meat, Hygienic handling and storage offish: chilling or icing, refrigeration, pest infestations in stored fishery products, natural means to control pests infestation, Food-grade coatings as processing aids, Integrated pest management.

Post-Harvest Management -II 3 (2+1)

Practical

- Study of physico-chemical properties of cereals, pulses and oils seeds,
- Determination of gluten content in wheat flour,
- Study of conditioning of wheat,
- Milling of wheat and rice by laboratory mill,
- Study of pre-treatment and milling of pulses,
- Study of oil expression equipment,
- Manufacture of value- added products, millets, guar gum and other minor crops,
- Study of primary processing of fish or meat,
- Study of preservation methods for fish/meat,
- Preparation of traditional value-added products from fish/meat,
- Preservation of meat/fish by freezing, Preservation of meat/fish by curing and pickling,
- Preservation of meat/fish by dehydration,
- Preparation of value-added poultry meat products,
- Quality evaluation and grading of eggs,
- Visit to processing facilities.

Standards and Certification for Natural Farming 3(2+1)

Objectives

1. Tolearn the natural farming standards and certification processes vis a vis other standards and certification in vogue.
2. To impart the methods of certification in natural farming based set of standard practices.
3. To understand the process of quality control of inputs and products in natural farming.

Theory

Scope and procedure of Natural Farming Certification System (NFCS), definitions, Crop production, habitat care, conversion, landscape, seeds and planting materials, diversity in crop production, inputs for soil and fertility management, Insect, pest and weed management, Contamination control, Transition / conversion Period and requirements for transition, Soil and water conservation, collection of non-cultivated material of plant origin forest produces, Livestock production, General requirements, Feed and fodder, Health care, Processing and handling, General requirement: Ingredients, Processing Methods, Packaging, Labelling, Processed Products, Storage and Transport, Marketing, Practices for use in soil preparation, Soil enrichment, practices for seed treatment, pest and disease management, procedure to evaluate natural farming input, Ethical Aspects — Animal welfare and socio-economic aspects, practices to be followed for livestock reared in natural farm, International and national regulations on quality assurance and certification for nonchemical farming systems, ICT enabled certification process.

Standards and Certification for Natural Farming 3(2+1)

Practical

- Visit to natural farm,
- Formulation of natural farm plan,
- Characterization of farm sources (Investigation) and filling up of application forms for certification,
- Exposure of students with the team certification agency,
- Group discussion on natural farm conversion period and standards requirements,
- Quality assurance and labelling requirement for certification of farm under natural farming standards.
- Visit of certification agency and learning about management of records and inspections.

Aqua-based Natural Farming3 (2+1)

Objectives

I. To acquaint students about natural farming and technologies pertaining to aquaculture and their relationship with farming activities

2. Basic understanding and knowledge on the overall view of the aquaculture sector, how to start aquaculture venture, opportunities and challenges

Theory

Physical, biological and components of aquatic ecosystem, Ecological structure and function of aquatic ecosystem, Organic matter recycling and nutrient cycles, Food web and food chain, Heterotrophic activities, Energy and matter flow in the ecosystem, Eutrophication and pollution. Definition and scope of aquaculture, Global and national scenario, Trends and constraints, Different aquaculture systems and utilization of natural food, pond, pen, enhanced fisheries, Pre- stocking, stocking and post-stocking interventions for primary fish survival and growth, Aquaculture and climate change, Natural productivity and carrying capacity, Secondary and tertiary production, Carrying capacity of aquaculture system and associated factors, Natural food and feeding behavior of fishes and shellfishes: Concept of ecosystem based aquaculture, Integration of crop, live- stock, horticulture and fish farming as complimentary activities, Comparative nutritional value and decomposition of different types of organic wastes on organic productivity, Nutrient values of common animal waste and agro residues as potential manures and biofertilizers, Definition of reservoirs in India: nature and extent of reservoirs, pen and cage culture in reservoir, Health management in aquaculture, Traditional techniques for curing fish diseases and pond management, Probiotics and bioremediation, Renewable and eco-friendly bio- inputs in aquaculture, Management and conservation natural aquatic ecosystems.

Aqua-based Natural Farming3 (2+1)

Practical

- Identification of important cultivable species,
- Estimation of carrying capacity, Practices on pre-stocking,
- Stocking and post stocking management,
- Growth studies in aquaculture system,
- Study on waste accumulation in aquaculture system (NH , Organic matter, C02),
- Analysis of certified manure/natural aquaculture inputs,
- Measurement of important soil and water quality parameters and their correction measures with certified standard materials/inputs,
- Preparation of artificial feeds using locally available permitted feed ingredients,
- Identification of different live food organisms and their rearing/culture with standard methods,
- Case studies on cage and pen culture.

Marketing of Natural Farming Produce 3 (2+1)

Objectives

1. To study about the specific concepts of marketing channels, price spread market segmentation and market margin
2. To develop skill on calculating market margin and farmers share in consumer's rupee in natural farming

Theory

Concepts and definition of market, Agricultural marketing, Marketing of natural farming products, Market structure, Marketing mix and market segmentation, Demand, supply and producer's surplus of agricultural commodities marketable and marketed surplus, Pricing consideration and approaches cost based and competition based pricing, Market promotion and publicity, Marketing management-segmentation, Targeting and positioning, Marketing function-physical function, facilitating functions, Market functionaries and marketing channels, Marketing efficiency, Marketing costs, Market margin and price spread, Role of government in marketing of natural farming products, Public sector institutions - CWC, SWC, FCI,CACP, APEDA, Risks in marketing-types of risks involved speculation and hedging, Forward markets and Future markets.

Marketing of Natural Farming Produce 3 (2+1)

Practical

- Study of relationship between market arrival and prices,
- Demand and supply curves,
- Calculation of marketable and marketed surplus,
- Identification of market channels for selected commodities,
- Computation of market margin,
- Price spread, visit to marketing institutions to study their organization and functioning.

Agronomic Practices 2 (1 +1)

Objectives

To acquaint students with management aspects of crops, weeds, water in natural farm fields To develop skills among students in practicing crop husbandry based natural farming

Theory

Classification of crops, Field crops — Origin, distribution, economic importance, soil and climatic requirements, varieties, cropping systems: definition, principles and its importance, physical resources, soil and water management in cropping systems, Importance of mixed cropping in natural farming, multiple cropping, alley cropping, sequential cropping and intercropping, cropping system indices - mechanism of yield advantage in intercropping systems, Complementary and competition relations, multi-storied cropping and yield stability in intercropping, Types of crops - Trap, cover, catch and restorative crops, ITK related to crops and cropping systems, Seeds- Traditional and recent varieties, classification of seeds, seed dormancy, nursery management, main field preparation, types of tillage, Seasons of India, Premonsoon sowing, Sowing techniques, Optimum time of sowing for different crops-ITK for seed treatment and seed selection, Methods of planting of crops,Time of planting of different crops and intercrops, Row spacing for different crops, Irrigation— Definition, irrigation types, water saving techniques and management, ITKS in irrigation and water management, Green manures and green leaf manures — Types of green manures, Ideal plant types for green manures, Nutrient content and biomass contribution, time and method of incorporation, decomposition pattern, Advantages and limitations, Weeds- Classification, habitat management of weeds, crop weed interaction, critical periods of weed competition, non-chemical weed management, weed mulch, trap crop, biological and herbal measures, Mulching — Types, cover crops — advantages and disadvantages, Soil moisture conservation approaches and water harvesting, ITKs on soil and water conservation. Alternate land use system — Definitions and types.

Agronomic Practices 2 (1 +1)

Practical

- Identification of different growth stages in crops, manures and seeds,
- Agriculture tools and implements, nursery and main field preparation, Animal drawn,
- Seed treatment and seed dormancy,
- Methods of sowing,
- Input requirement,
- Water requirement and water use efficiency (WUE),
- Weeds in wetland, irrigated up land, dryland system,
- Low cost technology, indices for cropping systems,
- Mulching practices,
- Green manuring practices,
- Alternate land use system,
- Wind break and shelter belt,
- ITKs in crops and agronomic practices.

Intellectual Property Rights1 (1+0)

Objectives

1. To acquaint the students about filing of patent, patent specification, patent claims, Patent opposition and revocation
2. To create awareness in the students about infringement, compulsory licensing, patent search and patent database

Theory

Introduction, importance and meaning of intellectual property, changing scenario ofagriculture in India, brief introduction to GATT, WTO,TRIPs and WIPO, Treaties for IPR protection: Madrid protocol, Berne Convention, Budapest treaty, etc.; Types of Intellectual Property and legislations covering IPR in India: Patents, Copyrights, Trademark, Industrial design, Geographical indications, Integrated circuits, Trade secrets, Patents Act, 1970 and patent system in India, patentability, process and product patent, filing of patent, patent specification, patent claims, Patent opposition and revocation, infringement, Compulsory licensing, Patent Cooperation Treaty, Patent search and patent database: Origin and history including a brief introduction to UPOV for protection of plant varieties, Protection of plant varieties under UPOV and PPV&FR Act of India, Plant Breeders rights, Registration of plant varieties under PPV&FR Act 2001, breeders, researchers and farmers rights, Traditional knowledge-Meaning and rights of TK holders. Convention on biological diversity, International treaty on plant genetic resources for food and agriculture (ITPGRFA), Indian Biological Diversity Act, 2002 and its salient features, access and benefit sharing. Understanding Intellectual Property Rights in ancient India.

Principles of Agricultural Economics and Farm Management 2 (2+0)

Objectives

To impart knowledge on judicious use of resources for optimum production

To impart knowledge about basic theories of demand, supply, production and exchange.

Theory

Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis: micro and macroeconomics, positive and normative analysis, Nature of economic theory: rationality assumption, concept of equilibrium, economic laws as generalization of human behavior, Basic concepts: Goods and services, desire, want, demand, utility, cost and price, wealth, capital, income and welfare, Agricultural economics: meaning, definition, characteristics of agriculture, importance and its role in economic development, Agricultural planning and development in the country, Demand: Meaning, law of demand, demand schedule and demand curve, determinants, utility theory; law of diminishing marginal utility, equi-marginal utility principle, Consumer's equilibrium and derivation of demand curve, concept of consumer surplus, Elasticity of demand: Concept and measurement of price elasticity, income elasticity and cross elasticity, Production: process, creation of utility, factors of production, input output relationship, Laws of returns: Law of variable proportions and law of returns to scale, Cost: Cost concepts, short run and long run cost curves, Supply: Stock v/s supply, law ofsupply, supply schedule, supply curve, determinants of supply, elasticity ofsupply, Distribution theory: meaning, factor market and pricing of factors of production, Concepts of rent, wage, interest and profit, National income: Meaning and importance, circular flow, concepts of national income accounting and approaches to measurement, difficulties in measurement, Population: Importance, Malthusian and Optimum population theories, natural and socio-economic determinants, current policies and program s on population control, Money: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, money supply, general price index, inflation and deflation, Economic systems: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning, Forms of business organizations, international trade and balance of payments, GST and its implication on Indian economy.

Semester VII

Community Mobilization for Natural Farming 2 (1 +1)

Objectives

1. To gain the knowledge about the fundamental concepts and principles of community mobilization, program planning and about groups 2. To develop an understanding of techniques and strategies for community mobilization in natural farming

Theory

Community mobilization — Meaning, historical background, types, strategies, steps, opportunities and challenges, Community mobilization process: community mobilization in rural areas and different models, community entry, profiling, social mapping, Yoga for community mobilization - Purpose, concepts, importance, types, benefits; SWOT Analysis- Meaning, advantages and case study, Social core value systems - Ethical principles, purpose, benefits, Development of understanding about rural society, Behavioral change and attitude, Program planning - Introduction, need and interest, concept, objectives and principles, process, Organizational structure - Define, key elements, types, span of control, Centralization and decentralization, formalization. Diffusion and Adoption Process- Define, meaning and stages. **Group:** Definition, types, stages of group formation, modes of group development, process for group formation - stages of SHG, FPO/FPC and FBO development, group decision making techniques, **FPO -** benefits, activities, implementing agencies to form and promote FPOs, procedure to form FPC and organizational structure, Conflict -Meaning, types, process, conflict resolution and management strategies.

Motivation: Definition, elements, individual and group needs, motivation process, Maslow's Hierarchy theory of needs and Adam's Equity theory, motivation cycle, classification of motives, types of motivation, techniques of motivation, role/importance/factors of motivation. Capacity building and institutionalizing mechanism, Counselling — Need, functions and types, Mentoring: Definition, process, philosophy and mentor-mentee relationship, Training - meaning, advantages, types and process, training methodologies for creating awareness about natural farming, Monitoring and Evaluation - Define, objectives, types, concept, difference, importance; Impact analysis on social, economic and environmental effects of natural farming. Case studies on community mobilization in natural farming, Traditional and social media for promoting natural farming, Government schemes and role of public and private institutions in promoting community mobilization in natural farming.

Community Mobilization for Natural Farming 2 (1 +1)

Practical

- Preparation of Community Resource Mapping in selected rural villages,
- Community needs assessment techniques — Participatory Rural Appraisal, Yoga, SWOT analysis, Organization involved in community mobilization,
- Program planning and organizing events,
- Formation and functioning of SHG, Establishment, organizational structure and promotion of FPO and FBO,
- Group decision making techniques - Conduct of focused group discussion,
- Conflict management techniques, Motivation — preparation of Maslow's hierarchy theory of needs assessment,
- Counselling and mentoring- conduct sessions among beneficiaries of natural farming community,
- Organizing, capacity building programs,
- Socio economic impact analysis on natural farming,
- Traditional and social media for promoting natural farming,
- Public and private institutions promoting community mobilization for natural farming.

Water Management 2 (1+1)

Objectives

1. To equip the students with the knowledge about the fundamental concepts and principles of water management
2. To develop an understanding of techniques and strategies for conserving water resources and improving water use efficiency both in rainfed and irrigated ecosystems

Theory

History of water management in ancient and medieval India, Rainfed and irrigated farming; Water management: Definition, scope and importance, Water resources: Use and over- utilization of surface and ground water, floods, drought, Dams-Benefits and problems, Region specific traditional water harvesting and management systems, Rainfed farming, Rainwater conservation technologies - In-situ and ex-situ storage, water harvesting and recycling, Rain water harvesting, Farm ponds, Ground water recharge, Contour and graded bunding, Chalkhal, Water budgeting, Irrigation water requirement, Micro irrigation: Sprinkler, Drip irrigation systems, pitcher irrigation system, maintenance of micro irrigation system, fertigation, advantages and limitations of fertigation, Watershed management - Concept, objectives, factors affecting and watershed planning, Reservoirs, Ground water harvesting and conservation (Tanka, Talai, Nadi, Nada, Talab, Khadin), Indian Traditional Knowledge in rain water management, soil and water conservation.

Water Management 2 (1+1)

Practical

- Measurement of soil moisture by different soil moisture measuring instruments,
- Measurement of irrigation water requirement,
- Determination of bulk density by field method,
- Determination of field capacity by field method,
- Determination of permanent wilting point,
- Case study on water budgeting,
- Study of different types of farm ponds,
- Estimation offarm pond storage capacity,
- Testing of irrigation water quality, EC, pH and TDS,
- Study of different components of sprinkler irrigation system,
- Study of different components of drip irrigation,
- Field visit to micro irrigation system,
- Maintenance of different components of micro irrigation systems,
- Survey of watershed resources,
- Field visit to watershed, Field visit to water harvesting structure.

Research Methodology and Ethics 2 (1 +1)

Objectives

1. To provide basic knowledge on research tools and techniques
2. To develop skills on on-farm research and ethics in general with specific reference to natural farming

Theory

Research Ethics: Introduction, Ethical ethos- Researcher's obligations and participant's rights, Research Ethics: Researcher-Participant, General Ethics, Ethical Issues in India, Ethics Committees. Experimental techniques: Research design, sampling, data collection, On-station experimentation, On-Farm experimentation, tabulation, Statistical tools and analysis, techniques for interpretation of data, Geo-referenced characterization: Questionnaire design principles, Questionnaire design for consumers of organic products, Questionnaire design for farmers and producers of organic products, Questionnaire design for processors/traders/exporters, Geo-spatial analysis and mapping of natural farms/ producers/traders/consumers. **Meta data analysis:** Concepts, Niche area and crops for natural farming: Parameters for niche area and crop, Different scales of niche area, Tools and steps in niche area and crop identification, Parameterization and classification based on macro, regional and micro level.

Climate resilience of natural farming: Methodology for identification of climate resilient production systems, GHGS estimation using IPCC, GHG's measurement using instrumentation, Global warming potential, Energy and Carbon budgeting. Bio-chemical and molecular signature of natural produces, Commercial project formulation on natural farming: Internal rate of return, Pay Back period, B:C ratio, Net Present Value, Model project formulation for organic farming, Impact analysis tools and methods, Farming system model development: Practical synthesis of natural farming system model, Estimation of GHG emission from natural farm using IPCC tools, Identification of niche area and crops for a district or block, Identification of climate resilient production system using long term meteorological data, Commercial project formulation, Geospatial analysis using GIS platform, Comparative carbon and energy budgeting between organic and natural farms, Comparative food quality between natural and conventional produces by metaanalysis.

Weather Forecasting 3 (2+1)

Objectives

1. To provide the knowledge on different types of weather forecasting
2. To acquaint the importance ofweather forecasting and its applications 3. To train the students on the ancient weather based agricultural practices

Theory

Ancient wisdom on weather forecasting: Rainfall prediction, analysis and forecast Of winter monsoon based on ancient literature and simulated models, Rain forecasting in India Almanacs (Panchangs), Measurement of rainfall, Testing of traditional methods of weather forecast, Ancient astronomers, Indian Almanac (Nakshatras, Rashi, Months, Paksha, Seasons, Tithi, Var, Yog, Karn, Nadi,), Krishi Panchang, Effect of planets on weather, Ancients methods of weather forecast (Analytical methods, Observational methods). Principles of Astro- Meteorology, Rainfall predictions techniques (1) Parashara technique, (2) Varahamihira technique, (3) Predictions based on planets, (4) Bio-indicators, (5) Other ancient rainfall predictions, Sudden rainfall, Indications of famine, The method of ascertaining the type of cloud of the year, Folklore regarding weather forecasting (Ghagh and his wife Bhaddri), (6) ITKS with description and the practitioner for weather forecasting, Meaning and scope of agricultural meteorology, Earth atmosphere-Its composition, extent and structure, Atmospheric weather variables, Atmospheric pressure, its variation with height, Nature and properties of solar radiation, solar constant, depletion of solar radiation, shortwave, longwave and the normal radiation, net radiation, albedo, Atmospheric temperature, temperature inversion, lapse rate, daily and seasonal variations of temperature, vertical profile of temperature, Energy balance of earth, Atmospheric humidity, concept of saturation, vapor pressure, process of condensation, formation ofdew, fog, mist, frost, cloud, Precipitation, process of precipitation, types of precipitation such as rain, snow, sleet and hail, cloud formation and classification, Artificial rain making. Wind: causes and types of wind, cyclone, anticyclone, land breeze and sea breeze, General circulation; Indian Monsoon - mechanism and importance in Indian agriculture, Weather forecasting - types of weather forecast and their uses, IT K in weather forecasting, Weather hazardsHigh winds, drought, floods, tornado, frost, tropical cyclones, thunderstorms, dust storm, lightning and hailstorms and extreme weather conditions such as heat-wave and cold-wave, Agriculture and weather relations, Modifications of crop microclimate, climatic normal for crop and livestock production, Climate change, climatic variability, global warming, causes of climate change and its impact on regional and national agriculture, Types of climate models — GCM and RCM.

Weather Forecasting 3 (2+1)

Practical

- Visit of Agrometeorological Observatory, site selection, layout, exposure to instruments and data acquisition techniques,
- Study and use of different instruments for measuring air temperature, rainfall,
- Estimation of net short-wave radiation, net long wave radiation, total global and net radiation,
- Measurement of albedo and sunshine duration, wind speed and direction,
- Measurement of evaporation rate using USWB Class A Open Pan Evaporimeter,
- Determination of relative humidity and vapour pressure, Study and use of Duvdevani DewGauge for measurement of Dew,
- Computation of climatic normal using historical weather data,
- Study of ancient methods of weather forecast, Preparation of Krishi Panchang, Testing local folklore for weather forecasting.

Basic and Applied Agricultural Statistics 3 (2+1)

Objectives

1. To provide an idea on Statistical concepts of both descriptive and inference Statistics
2. To impart knowledge about tests of significance, ANOVA and sampling theory

Theory

Introduction to Statistics and its applications in agriculture, Types of Data, Scales of measurements of Data, Summarization of Data, Classification of Data, Frequency Distribution, Methods of Classification, Definition of Grouped and Ungrouped Data, Definition of Class Interval (formula for determining the no. of class interval), Width of CI, Class Limits (Boundaries), Mid Points, Types of Frequency Distribution, Diagrammatic Presentation of Data, Bar Diagrams — Simple, Multiple, Sub-divided and Percentage Bar Diagrams, Pie-diagram, Graphical Presentation of Data — Histogram, Frequency Polygon and Ogives.

Measures ofCentral Tendency, Requisites for an Ideal Measure of Central Tendency, Different Types of Measure, Arithmetic Mean— Definition, Properties, Merits, Demerits and Uses, A.M. (examples) for Grouped and Ungrouped Data, Step-deviation Method, Weighted Mean, Definition of Geometric Mean and Harmonic Mean, Relationship between A.M., G.M. and H.M. MedianDefinition, Merits, Demerits and Uses, Graphical Location of Median, Mode- Definition, Merits, Demerits and Uses, Graphical Location of Mode, Relationship between Mean, Median and Mode, Measures of Dispersion, Characteristics for an Ideal Measure of Dispersion, Different Types of Measures of Dispersions, Definition of Range, Interquartile Range, Quartile Deviation and Mean Deviation, Standard Deviation- Definition, Properties, S.D. and Variance for Grouped and Ungrouped Data, Variance of Combined Series, Coefficients of Dispersions. Co-efficient of Variation, Measures of Skewness and Kurtosis: Definition of Symmetrical Distribution, Definition of Skewness, Measures of Skewness, Definition of Kurtosis, Measure of Kurtosis, Relationship between Mean, Median and Mode for Symmetrical and Skewed Distribution.

Basic and Applied Agricultural Statistics 3 (2+1)

Theory Cont---

Probability Theory and Normal Distribution, Introduction to Probability, Basic Terminologies,
Classical Probability- Definition and Limitations, Empirical Probability- Definition and Limitations, Axiomatic Probability, Addition and Multiplication Theorem (without proof), Conditional Probability, Independent Events, Simple Problems based on Probability, Definition of
Random Variable, Discrete and Continuous Random Variable, Normal Distribution- Definition, Prob. Distribution, Mean and Variance, Assumptions of Normal Distribution, Normal Probability Curve. Correlation and Regression, Definition of Correlation, Scatter Diagram, Karl Pearson's Coefficient of Correlation, Types of Correlation Coefficient, Properties of Correlation Coefficient, Definition of Linear Regression, Regression Equations, Regression Coefficients, Properties of Regression Coefficients.
Tests of Significance, Definition, Null and Alternative Hypothesis, Type I and Type Il Error, Critical Region and Level of Significance, One Tailed and Two Tailed Tests, Test Statistic, One Sample, Two Sample and Paired t-test with Examples. F-test for Variance.
ANOVA and Experimental Designs, Definition of ANOVA, Assignable and Non assignable Factors, Analysis of One-way Classified Data, Basic Examples of Experimental Designs, Terminologies, Completely Randomized Design (CRD).
Sampling Theory, Introduction, Definition of Population, Sample, Parameter and Statistic, Sampling Vs Complete Enumeration, Sampling Methods, Simple Random Sampling with Replacement and without Replacement, Use of Random Number Table.

Practical

- Diagrammatic and Graphical representation of data,
- Calculation of A.M., Median and Mode (Ungrouped and Grouped data),
- Calculation of S.D. and CV. (Ungrouped and Grouped data),
- Correlation and Regression analysis,
- Application of t-test (one sample, two sample independent and dependent),
- Analysis of variance one way classification,
- CRD,
- Selection of random sample using simple random sampling.

Semester VIII

	Course Title	Credit Load
	For students opting for 4-year B. Sc. Ag. (Hons.) Natural Farming degree Based on the theme Of Student READY Program, students may be allotted Experiential Learning/Hands on Training/Skill development/ RAWE/ Industrial attachment/IPT/ student project and Internship depending upon the resources and priorities.	20 Credits (Minimum20 weeks)
	RAWE	8
	Experiential Learning	6
	IPT/Industrial Attachment	4
	Student Project/Internship	2

For B.Sc. Ag. (Hons.) Natural Farming

For B.Sc. Ag. (Hons.) Natural Farming

- Project work: Student Project will be either R&D based, field study based (RAWE) or entrepreneurship based (incubation/experiential learning). The project work will be of above mentioned credits (Minimum 20 weeks) during which they will be attached to projects to learn, field experience, laboratory and field research or incubation centres or experiential learning units for developing viable projects.
- Internship work: The students will be attached to industry/ research institute/ commercial farm/ NGO/ agribusiness entrepreneur for 10 weeks with 10 credits to complete internship work and to learn entrepreneurship skills.
- Each student will submit individual report for both project work and internship work which will be evaluated and given grades.

ELECTIVE COURSES

	Course Number	Title
1.	Ag Econ (E) 413	Agri-Business Management
2.	Agron (E) 413	Management of Natural Resources
	Course Number	Title
3.	EE (E) 413	Agricultural Journalism
4.	Hort (EA) 323	Landscaping
5.	GPB (E) 413	Commercial Plant Breeding
6.	Hort (EB) 323	Food Safety and Standards
7.	PP (EA) 322	Bio Formulation and Nano formulation
8.	PP (EB) 413	Biopesticides and Biofertilizers
9.	Ag Met (EA) 413	System Simulation and Agro-advisory
10.	Hort (EC) 413	Hi-tech Horticulture
11.	Hort (ED) 413	Protected Cultivation
12.	Ag Met (EB) 413	Climate Resilient Agriculture
13.	ABT(E) 413	Biotechnology of Crop Improvement
14.	Soil (E) 412	Geoinformatics and Remote Sensin , Precision Farmin
15.	Hort (EE) 413	Micro-propagation Technologies
16.	SS%413	Commercial Seed Production

- More electives to be included by the universities / institutions, based on the facilities available, such as nano formulations, drones use in agriculture etc

Agri-Business Management 3 (2+1)

Objective

To impart knowledge on understanding the concepts processes, significance and role of management and organizational behaviour

Theory

Transformation of agriculture into agribusiness, various stakeholders and components of agribusiness systems, Importance of agribusiness in the Indian economy and New Undergraduate Degree Program in Natural Farming
Agricultural Policy, Distinctive features of Agribusiness Management: Importance and needs of agro-based industries, Classification of industries and types of agro-based industries, Institutional arrangement, procedures to set up agro-based industries, Constraints in establishing agro-based industries, Agri-value chain: Understanding primary and support activities and their linkages, Business environment: PEST and SWOT analysis, Management functions: Roles and activities, Organization culture, Planning, meaning, definition, types of plans, Purpose or mission, goals or objectives, strategies, polices procedures, rules, programs and budget, Components of a business plan, Steps in planning and implementation, Organization staffing, directing and motivation, Ordering, leading, supervision, communications, control, Capital management and Financial management of Agribusiness, Financial statements and their importance, Marketing Management: Segmentation, targeting and positioning, Marketing mix and marketing strategies, Consumer behavior analysis, Product Life Cycle (PLC), Sales and Distribution management, Pricing policy, various pricing methods, Project management: Definition, project cycle, identification, formulation, appraisal, implementation, monitoring and evaluation, Project appraisal and evaluation techniques.

Agri-Business Management 3 (2+1)

Practical

- Study of agri—input markets: Seed, fertilizers pesticides,
- Study ofoutput markets: grains, fruits, vegetables, flowers,
- Study of product market, retails trade commodity trading, and value added products,
- Study of financing institutions- Cooperative, Commercial Bank, RRBs, Agribusiness Finance Limited, NABARD,
- Preparations of projects and Feasibility reports for agribusiness entrepreneur,
- Appraisal/ evaluation techniques of identifying viable project- Non-discounting techniques,
- Case study of agro- based industries,
- Trend and growth rate of price of agricultural commodities,
- Net present worth technique for selection of viable project,
- Internal rate of return,

Management of Natural Resources 3 (2+1)

Objective

1. To enlighten students about available natural resources and their relationship with crop production
2. To impart the knowledge of principles and practices of natural resource management

Theory

Introduction to Natural Resource Bases: Concept of resource, classification of natural resources, Factors influencing resource availability, distribution and uses, Interrelationships among different types of natural resources, Concern on productivity issues, Ecological, social and economic dimension of resource management, Land resources: Land as a resource, Dry land, land use classification, land degradation, man induced landslides, soil erosion and desertification, Landscape impact analysis, wetland ecology and management, Water resources: Use and overutilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems, Water ecology and management, Energy resources: Growing energy needs, renewable and non- renewable energy sources, use of alternate energy sources, Resource Management Paradigms: Resource management the evolution and history of resource management paradigms, Resource conflicts: Resource extraction, access and control system, Approaches in Resource Management: Ecological approach; economic approach; ethnological approach; implications of the approaches: integrated resource management strategies, Introduction to soil and water conservation and causes of soil erosion, Definition and agents of soil erosion, Water erosion: Forms of water erosion, gully classification and control measures, Soil loss estimation by universal soil loss equation: Soil loss measurement techniques. Principles of erosion control: Introduction to contouring, strip cropping, Contour bund: Graded bund and bench terracing, Wind erosion: Mechanics of wind erosion, types of soil movement, Principles of wind erosion control and its control measures, Water harvesting techniques: Lining of ponds, tanks and canal systems.

Management of Natural Resources 3 (2+1)

Practical

- Identifying natural resources and their utility,
- Practicing survey: Principles and educating to use pacing technique for measurement,
- Area calculations through chain survey.
- GPS demo for tracking and area measurement,
- Estimation of soil loss and calculation of erosion index,
- Leveling concepts and practical utility in agriculture,
- Preparation of contour maps,
- Concept of vegetative water ways and design of grassed water ways,
- Wind erosion and estimation process,
- Different irrigation pumps and their constructional differences,
- Farm pond construction and its design aspects, Visit to nearby farm pond,
- Visit to an erosion site, Exposure to strip cropping/contour bunding.

Agricultural Journalism 3 (2+1)

Objective

To acquaint the qualities and role of journalist /journalism in agricultural development and development of media

Theory

Journalism — Meaning, nature, importance and types of journalism, Agricultural Journalism — Meaning, definition, principle, objectives, types, and scope, Similarities and difference between agricultural journalism and other types of journalism, Role of agricultural journalist, Training of agricultural journalist, Qualities of journalist, Role of journalist /journalism in agricultural development and development of newspaper and magazines readers, Newspaper and magazines as communication media: Characteristics, kinds and functions of newspaper and magazines, Characteristics of newspaper and magazines readers, Form, content, style and language of newspaper and magazines, Standard part of newspaper and magazines, The agricultural story: Types of agriculture stories, subject matter of the agricultural story, structure of the agricultural story, Gathering farm information - Sources of farm information: abstracting from research and scientific materials, interviews, coverage of events, other sources: electronic media, field study. Success stories-definition, nature, components, guidelines of writing a success story, Writing a news story difference between news and feature story, the principle of writing a news story, Inverted pyramid structure, Organizing the material, treatment of the story, writing the news lead and the body Readability measure-readability ease score, automated readability index, gunning fog index, How to improve readability of articles and stories, Use of photograph in agricultural journalism-Basic principles of photography — composition, exposure, lens, light Use of artwork (Graphs, charts maps, etc), Writing the captions, Editorial mechanism: Copy reading, headline and title writing Proof reading: definition, signs and symbols of proofreading, level of proofreading, duties of a proof-reader, Layout— Meaning, principles of layout and design.

Agricultural Journalism 3 (2+1)

Practical

- Practice in writing an agricultural news story,
- Practice in writing an agricultural feature story,
- Covering agricultural events for the information collection,
- Practice in interviewing for the information collection,
- Abstracting stories from research and scientific materials and wire services,
- Selecting pictures and artwork for the agricultural story,
- Practice in editing, copy reading, Practice in headline and title writing,
- Practicing proof reading,, Practice in lay outing of newspaper,
- Testing copy with a readability formula,
- Visit a publishing office.

Landscaping 3 (2+1)

Objectives

I. To educate the students on designing different styles and types of gardens

2. To enable the students to identify different ornamental plants and their utilization in landscaping design

3. To enable students to design landscapes in softwares like AUTOCAD, ARCHCADE etc.

Theory

Importance and scope of landscaping, Principles of landscaping, garden styles and types terrace gardening, vertical gardening, garden components, adornments, lawn making, rockery water garden, walk-paths, bridges, other constructed features etc. gardens for special purposes, Trees: Selection, propagation, planting schemes, canopy management, shrubs and herbaceous perennials: selection, propagation, planting schemes, architecture, Climber and creepers importance, selection, propagation, planting, Annuals: selection, propagation, planting scheme, Other garden plants: palms, ferns, grasses and cacti succulents, Pot plants: selection, arrangement, management, Bio-aesthetic planning: Definition, need, planning; landscaping of urban and rural areas, Peri-urban landscaping, Landscaping of schools, public places like bus station, railway station, townships, river banks, hospitals, play grounds, airports, industries, institutions, Bonsai principles and management, Lawn: establishment and maintenance. CAD application.

Landscaping 3 (2+1)

Practical

- Identification of trees, shrubs, annuals, pot plants;
- Propagation of trees, shrubs and annuals,
- Care and maintenance of plants,
- Potting and repotting,
- Identification of tools and implements used in landscape design,
- Training and pruning of plants for special effects,
- Lawn establishment and maintenance,
- Layout of formal gardens, informal gardens, special type of gardens (sunken garden, terrace garden, rock garden) and designing of conservatory and lathe house.
- Use of computer software, visit to important gardens [parks /institutes.

Commercial Plant Breeding 3 (2+1)

Objectives

1. To discuss about hybrid development and various crop improvement aspects of field crops viz., rice, wheat, maize, pearl millet, sorghum, pigeonpea, chickpea, green-gram, black gram, lentil, soybean, groundnut, rapeseed- mustard, cotton etc.
2. To provide understanding on tissue culture and biotechnological approaches as alternative strategies for development of line and cultivars
3. To impart knowledge on seed production, release and notification of varieties and PPV&FR Act, 2001

Theory

Types of crops and modes of plant reproduction, Line development and maintenance breeding in self- and cross- pollinated crops (A/B/R and two line system) for development of hybrids and seed production, Genetic test of commercial hybrids, Advances in hybrid seed production of maize, rice, sorghum, pearl millet, castor, sunflower, cotton pigeon pea, Brassica etc. Quality seed production of vegetable crops under open and protected environment, Alternative strategies for the development of the line cultivators: haploid inducer, tissue culture techniques and biotechnological tools, IPR issues in commercial plant breeding: DUS testing and registration of verities under PPV and FRA Act, Variety testing, release and notification systems in India, Principles and techniques of seed production, types of seeds, quality testing in self and cross pollinated crops.

Commercial Plant Breeding 3 (2+1)

Practical

- Floral biology in self and cross pollinated species,
- Selfing and crossing techniques,
- Techniques of seed production in self and cross pollinated crops using A/B/R and two line system,
- Learning techniques in hybrid seed production using male- sterility in field crops,
- Understanding the difficulties in hybrid seed production,
- Tools and techniques for optimizing hybrid seed production,
- Concept of rouging in seed production plot,
- Concept of line its multiplication and purification in hybrid seed production,
- Role of pollinators in hybrid seed production,
- Hybrid seed production techniques in sorghum, pearl millet, maize, rice, rapeseed-mustard, sunflower, castor, pigeon pea, cotton and vegetable crops,
- Sampling and analytical procedures for purity testing and detection of spurious seed, Seed drying and storage structure in quality seed management,
- Screening techniques during seed processing viz. grading and packaging, Visit to public and private seed production and processing plants.

Food Safety and Standards 3 (2+1)

Objectives

1. To develop the skills to convert raw materials into safe, attractive food products
2. To manage the production of food products
3. To use scientific knowledge to develop new products

Theory

Food safety —Definition, Importance, Scope and Factors affecting Food Safety, Hazards and Risks, Type of Hazards - Biological, Chemical Physical hazards, Management of hazards — Need, Control of Parameters, Temperature Control, Food Storage, Production Design, Hygiene and Sanitation in Food Service Establishments- Introduction. Sources of contamination and their control, Waste Disposal, Pest and Rodent Control, Personnel Hygiene, Food safety Measures, Food safety Management Tool- Basic concepts, PRPs, GHPs, GMPs, SSOPs etc. HACCP.ISO series, TQM- Concept and need for quality, components of TQM, Kaizen, Risk Analysis, Accreditation and Auditing, Water Analysis, Surface Sanitation and Personal Hygiene, Food laws and Standards Indian Food Regulatory Regime, FSSAI. Global Scenario CAC, Other laws and standards related to food, Recent Concerns-New and Emerging Pathogens, Packaging, Product labelling and nutritional labelling, Genetically modified food/transgenic, Organic foods, Newer approaches to food safety, Recent Outbreaks, Indian and International Standards for food products.

Practical

- Water quality analysis: physio—chemical and microbiological,
- Preparation of different types of media, Microbiological examination of different food samples,
- Assessment of surface sanitation by swab/rinse method,
- Assessment of personal hygiene.
- Biochemical tests for identification of bacteria,
- Scheme for the detection of food borne pathogens,
- Preparation of plants for implementation of FSMS- HACCP, ISO: 22000.

Bioformulation and Nano-formulation 3 (2+1)

Objectives

1. To enable students to acquire expertise and skill to develop bioformulation and Nanoformulation
2. To know the importance of biopesticides and biofertilizers
3. To make the students know about various techniques involved in biofertilizers and biopesticides production
4. To get knowledge on essential oils, botanicals, predators, parasitoids, Pheromones, and parapheromone and their application in insect-pest management
5. To get concepts on agrochemical formulations with nanoparticles and acquaint them with nanotechnology

Theory

Introduction and history of biological control of pests and diseases, Microbial biopesticides: the global and Indian market scenario; biopesticides for organic agriculture, Different phytopathogenic biocontrol agents: Mode of action, Different entomopathogenic biocontrol agents: Mode of action, Microbial inoculants as biofertilizer candidates, Production, quality assessment and methods of application of biopesticides and biofertilizers, Regulatory system of biopesticides in India, Formulations of plant essential oils, botanicals, Pheromone, and parapheromone and their application in insect pest management, Use of predators and parasitoids for insect pest management, Nanotechnology: Its applications in pest and disease diagnosis and management, Nano biopesticides: Concept and importance, different techniques of producing nano biopesticides, Nano fertilizers: Concept and importance, Types of nano fertilizers, Different techniques of producing nano fertilizers, Green synthesis of nano fertilizers, green slow-release fertilizer composition based on urea-modified hydroxyapatite nanoparticles

Bioformulation and Nano-formulation 3 (2+1)

Practical

- Introduction and acquaintance with biopesticide laboratory,
- Preparation of culture media,
- Isolation and purification of bioagent from soil and infected insects, Microscopic study of different microbial bioagents,
- In vitro assay of microbial bioagents against plant pathogens.
- In vitro compatibility study among different microbial bioagents, Mass multiplication of biopesticides,
- Population enumeration of biocontrol agents in different biopesticides,
- Preparation of plant extracts and their efficacy test against insect pests,
- Use of pheromone para pheromone for monitoring and management of insect pests,
- Bioassay of Entomopathogenic biocontrol agents on insect-pests,
- Preparation of microbial inoculants of biofertilizer microbes,
- Compatibility of biofertilizer microbes,
- Preparation of solid and liquid consortia of biofertilizer microbes.

Biopesticides and Biofertilizer3 (2+1)

Objectives

1. To provide knowledge on principles, methods, and mechanisms of bio-control agents and their use against plant diseases
2. To provide knowledge on principles, methods, and mechanism ofbiofertilizers and their use in agriculture

Theory

History and concept of bio pesticides, Importance, scope and potential of biopesticides, Definitions, concepts and classification of biopesticides viz. Pathogen, botanical pesticides, and bio rationales. Botanicals and their uses. Mass production technology of bio-pesticides, Virulence, pathogenicity and symptoms ofentomopathogenic pathogens and nematodes, Methods ofapplication of biopesticides. Methods of quality control and Techniques of bio pesticides. Impediments and limitation in production and use of biopesticides.
Biofertilizers - Introduction, status and scope. Structure and characteristics features of bacterial biofertilizers — Azospirillum, Azotobacter, Bacillus, Pseudomonas, Rhizobium and Frankia; Cynobacterial bio fertilizers- Anabaena, Nostoc, Hapalosiphon and fungal biofertilizers — AM mycorrhiza and ectomycorhiza. Nitrogen fixation —Free living and symbiotic nitrogen fixation. Mechanism of phosphate solubilisation and phosphate mobilization, K solubilisation. Production technology: Strain selection, sterilization, growth and fermentation, mass production of carrier based and liquid biofertizers, FCO specifications and quality control of biofertilizers. Application technology for seeds, seedlings, tubers, sets etc. Biofertilizers: Storage, shelf life, quality control and marketing. Factors influencing the efficiency of biofertilizers.

Biopesticides and Biofertilizer3 (2+1)

Practical

- Isolation and purification of important biopesticides: Trichoderma, Pseudomonas, Bacillus, Metarhyzium etc. and its production,
- Identification of important botanicals,
- Visit to biopesticide laboratory in nearby area,
- Field visit to explore naturally infected cadavers,
- Identification of entomopathogenic entities in field condition,
- Quality control of biopesticides,
- Isolation and purification of Azospirillum, Azotobactor, Rhizobium, P-solubilizers and cyanobacteria, Mass multiplication and inoculums production of biofertilizers,
- Isolation of AM fungi- Wet sieving method and sucrose gradient method,
- Mass production of AM inoculants.

System Simulation and Agro-advisory 3 (2+1)

Objectives

I. To impart the knowledge of statistical and simulation modelling in crop yield estimation

2. To get acquainted with different weather forecasting techniques and their usability analysis

3. To study about the preparation and dissemination of agro-advisory bulletin

Theory

System Approach for representing soil-plant-atmospheric continuum, system boundaries, Crop models, concepts and techniques, types of crop models, data requirements, relational diagrams, Evaluation of crop responses to weather elements; Elementary crop growth models: Calibration, validation, verification and sensitivity analysis, Potential and achievable crop production- concept and modelling, techniques for their estimation, Crop production in moisture and nutrients limited conditions; components ofsoil water and nutrients balance, Weather forecasting, types methods, tools and techniques, forecast verification; Value added weather forecast, ITK for weather forecast and its validity; Crop- Weather Calendars; Preparation of agro-advisory bulletin based on weather forecast, Use of crop simulation model for preparation of Agro-advisory and its effective dissemination.

System Simulation and Agro-advisory 3 (2+1)

Practical

- Preparation of crop weather calendars,
- Preparation of agro-advisories based on weather forecast using various approaches and synoptic charts,
- Working with statistical and simulation models for crop growth,
- Potential and achievable production; yield forecasting, insect and disease forecasting models,
- Simulation with limitations of water and nutrient management options,
- Sensitivity analysis of varying weather and crop management practices,
- Use of statistical approaches in data analysis and preparation of historical, past and present meteorological data for medium range weather forecast,
- Feedback from farmers about the agro-advisory.

Hi-tech Horticulture 3 (2+1)

Objective

Theory

Introduction and importance; Nursery management and mechanization; micro propagation of horticultural crops; Modern field preparation and planting methods, Protected cultivation: advantages, controlled conditions, method and techniques, Micro irrigation systems and its components; EC, pH based fertilizer scheduling, canopy management, high density orcharding, Components of precision farming : Remote sensing, Geographical Information System (GIS), Differential Geo- Positioning System (DGPS), Variable Rate Applicator (VRA), application of precision farming in horticultural crops (fruits, vegetables and ornamental crops); mechanized harvesting of produce.

Practical

- Types of polyhouses and shade net houses,
- Intercultural operations, tools and equipment's identification and application,
- Micro propagation,
- Nursery- Portrays, micro-irrigation,
- EC, pH based fertilizer scheduling,
- Canopy management,
- Visit to hi-tech orchard/nursery.

Protected Cultivation 3 (2+1)

Objectives

1. To educate the students on different protected structures
2. To educate the students on agro-techniques and management of different horticultural crops under protected environmental conditions

Theory

Protected cultivation: Importance and scope, status of protected cultivation in India and World types of protected structure based on site and climate, Cladding material involved in greenhouse/ poly house, Greenhouse design, environment control, artificial lights, Automation, Soil preparation and management, Substrate management, Types of benches and containers, Irrigation and fertigation management, Propagation and production ofquality planting material of horticultural crops, Greenhouse cultivation of important horticultural crops-rose, carnation, chrysanthemum, gerbera, orchid, anthurium, lilium, tulip, tomato, bell pepper, cucumber, strawberry, pot plants etc. Cultivation of economically important medicinal and aromatic plants, Off- season production of flowers and vegetables, Insect pest and disease management.

Practical

- Raising of seedlings and saplings under protected conditions,
- Use of portrays in quality planting material production,
- Bed preparation and planting of crop for production,
- Inter cultural operations,
- Soil EC and pH measurement.
- Regulation of irrigation and fertilizers through drip, fogging and misting.

Climate Resilient Agriculture 3 (2+1) ctives

Objectives

1. To impart the concept of climate resilient agriculture under the present context of climate change
2. To study the integrated role of different sectors in building resilience to climate change in agriculture

Theory

Climate change and impacts of climate change on agriculture and food security; crop productivity under different climate change scenarios including extreme events such as drought, flood, pest and disease outbreak etc. Basics of adaption and mitigation in the agricultural sectors; analyzing and assessing climate vulnerability to identify vulnerable sectors and possible adaptation options in agriculture; assessing biophysical and socio- economic impacts on agricultural sector; risk assessment strategies, preparedness for weather and climate risks in agriculture; application of geospatial tools and techniques for sustainable agriculture.

Climate resilient agriculture (CRA) — Concept, scope and importance with special reference to India, climate resilient technologies for enhancing crop productivity and sustainability, Role of weather and climatic information, agro-advisories, ICTs and simulation models; climate resilient agronomic practices — Crop/cultivar selection, crop diversification/ crop mixtures; Water management practices — Rain water harvesting, micro-irrigation, deficit irrigation and drainage management, organic/natural farming, integrated farming systems (IFS); site specific nutrient management (SSNM), conservation agriculture technologies to build soil organic carbon, harnessing microbial biodiversity, biomass recycling; use of renewable sources of energy; climate resilient pest-disease management strategies.

Breeding strategies for development ofclimate change resilient crops and varieties, development of biotic and abiotic stress tolerant/resistant cultivars under changed climatic scenarios including extreme weather events.

Climate Resilient Agriculture 3 (2+1)

Practical

- Acquaintance with meteorological instruments including AWS,
- Statistical techniques to study trend of climatic parameters,
- Analysis of extreme weather events using non- parametric tests,
- Building climate change scenarios under different futuristic emission of GHGs,
- Designing strategies to mitigate the effect of climate change using climate resilient crops/cultivars,
- Climate resilient technologies and manipulation of cropping patterns, Acquaintance with ICTs for effective dissemination of local weather information and agro- advisories,
- Analysing carbon sequestration potential of different agro-ecosystems;
- Designing climate smart village model considering the availability of resources.
- Awareness program on climate change and climate resilient agriculture among farming community.

Biotechnology of Crop Improvement 3 (2+1)

Objectives

1. To acquaint with biotechnological tools of crop improvement
2. To know about direct and indirect methods of gene transfer
3. To introduce about gene editing in plants
4. To provide knowledge about marker assisted breeding and genomic selection

Theory

Impact of biotechnology on crop improvement and the perspective of society: Various biotechnological techniques available for crop improvement — Plant Tissue Culture, Genetic Engineering, Genome editing, Marker Assisted breeding and Genomic Selection, Biosafety regulations and their application in Agricultural Biotechnology, Soma clonal variation and its use in crop improvement: embryo culture, anther/pollen culture, somatic embryogenesis, artificial seeds, techniques of protoplast culture, regeneration and somatic cell hybridization, achievements and limitations, utility in the improvement of crop plants, Direct and Indirect methods of gene transfer in plants - Agrobacterium-mediated gene transfer in dicots and monocots, Direct DNA delivery methods (microinjection, particle gun method, electroporation), gene targeting, Gene silencing techniques, introduction to siRNA, siRNA technology, Micro RNA, construction of siRNA vectors, principle and application of gene silencing, creation of transgenic plants, debate over GM crops, introduction to methods of genetic manipulation in different model systems.

Introduction to genome editing: Various tools of genome editing, CRISPR-Cas9 with specific emphasis on Indian regulations, Cloning genomic targets into CRISPR/ Cas9 plasmids, electroporation of Cas9 plasmids into cells, purification of DNA from Cas9 treated cells and evaluation of Cas9 gene editing, in vitro synthesis of single guide RNA (sgRNA), using Cas9/sgRNA complexes to test for activity on DNA substrates, evaluate Cas9 activity by T7E1 assays and DNA sequence analysis, Applications of CRISPR/cas9 technology in crop plants, Marker Assisted Breeding and Genomic Selection: Introduction to various DNA- based markers and their use in marker-assisted breeding, Foreground Selection, Recombinant Selection and background Selection, Marker-assisted backcross breeding, marker-assisted selection — success stories, Introduction to Genomic Selection.

Practical

- Agrobacterium-mediated transformation in tobacco — Preparation of construct, transfer to binary vector,
- Transform Agrobacterium, prepare explant,
- Inoculation and Co-cultivation, antibiotic based selection of putative transformants,
- Validation using PCR;
- Genome editing- Preparation of CRISPR/Cas construct, direct transfer to plant,
- Analysis of the targets,
- Planning of a MABB program — Selection of parents,
- Crossing strategies, marker analysis.

Geoinformatics and Remote Sensing, Precision Farming 3 (2+1)

Objective

To provide a comprehensive knowledge of remote sensing, precision farming and its benefits in improving crop production and soil health management.

Theory

Introduction and history of remote sensing; sources, principles of remote sensing, propagation of radiations in atmosphere; Interaction with matter, application of remote sensing techniques land use soil surveys, crop stress and yield forecasting, Advantages and disadvantages of remote sensing, Remote sensing institutes in India, Basic concepts about geoinformatics.

Artificialintelligence: History ofartificial intelligence, Fundamentals of Big data and machine learning (ML), Use of artificial intelligence in autonomous systems: agricultural robots and drone monitoring systems, driverless tractors, automated sprinklers and self- harvesting machines etc.; Use of Al in crop analysis: monitoring soil quality, promoting organic crops, monitoring weeds, precision agriculture, using drones for crop analysis; Role of Al for sustainability and climate change, yield and demand forecasting, food tech/ wider value chain including impact of blockchain, Al use for in the emerging markets; Technology deployment like sensors, Al and agricultural technologies and How to scale Al for agricultural technologies applications, Responsible Al in agriculture, Data sharing; Expert System: Introduction to expert system, Characteristics and features of expert system,

Geoinformatics and Remote Sensing, Precision Farming 3 (2+1)

Practical

- Applications of Expert System, Importance of Expert systems, Rule based system architecture; Software Agents.1 321
- Familiarization with different remote sensing equipments and data products, Interpretation of aerial photographs and satellite data for mapping of land resources,
- Global positioning system (GPS), Basics of Geographic Information System (GIS), Georeferencing of topo sheets.
- Live examples and case study of Al use in Agriculture,
- Search and Control strategies: Blind search, Breadth - first search, Depth First search,
- Hill climbing method, Best First search, Branch and Bound search,
- Programming in Prolog Syntax and meaning of Prolog Programs,
- Using Data Structures, Controlling Back- tracking,
- Input and Output, Built-in Predicates, Using Prolog Grammar Rules,
- Higher level assignments/exercises for implementation using Prolog.

Micro-propagation Technologies 3 (2+1)

Objective:

To educate the students in detail about the sterilization techniques for explants, preparation of stocks and working solution, culturing of explants, regeneration of whole plants from different explants and hardening procedures

Theory

Introduction, History, Advantages and limitations. Types of cultures (seed, embryo, organ, callus, cell), Stages of micro propagation, Axillary bud proliferation (Shoot tip and meristem culture, bud culture), Organogenesis (callusy and direct organ formation), Somatic embryogenesis, cell suspension cultures, production ofsecondary metabolites, Soma clonal variation, Cryopreservation.

Practical

- Identification and use ofequipment's in tissue culture Laboratory,
- Nutrition media composition, sterilization techniques for media, containers and small instruments, sterilization techniques for explants,
- Preparation of stocks and working solution,
- Preparation of working medium,
- Culturing of explants: Seeds, shoot tip and single node, Callus induction, Induction of somatic embryos regeneration of whole plants from different explants,
- Hardening procedures.

Commercial Seed Production 3 (2+1)

Objective:

- To introduce the basic principles of planting material production at commercial scale and seed quality evaluation

Theory

General Principles of Seed Production: Raising the seed crop, Introduction, Procurement of a class of Improved seeds, Reporting to Monitoring or certification Agency, Principles and practices of selection of area and agronomic requirement of seed production of field crops, Importance of isolation distance and rouging, Principles of hybrid seed production in field crops, Principles and practices of selection of area and agronomic requirement of seed production of horticultural crops, Concept ofapomixis, male sterility and self-incompatibility and its application in hybrid seed production of horticultural crops.

General Principles of Seed Processing: Introduction, Objectives of Seed Processing, Seed Drying, Principles of Drying, Water vapour equilibrium, Methods of drying seeds, Cleaning and grading, Air and screen machines, Dimensional separators, Density separators, Surface texture separators, Colour separators, Spiral separators, Electric separators, Vibrator separators, Separation based on Affinity to liquids, Seed treatment, Temperature treatment, Chemical treatment, Bagging and Labelling

General Principles of Seed Testing: Seed testing - introduction, Procedure of Seed testing, components of seed quality testing genetic, physical, physiological and seed health testing, Seed sampling, Types of seed sampling, Requirements of sampling, Concept of seed viability and vigour; dormancy, types and principles of seed dormancy, Physiological quality of seed, Principles of seed germination, types of germination, biochemical and genetic basis.

Seed Certification: History, concept and objectives of seed certification; seed certification agency/organization and staff requirement Indian Minimum Seed Certification Standards (I.M.S.CS.) - General and specific crop standards including GM varieties, field and seed standards.

Commercial Seed Production 3 (2+1)

Seed Industry and Seed Marketing : Introduction, Evolution of the seed industry, Development of the vegetable and Flower seed industry, Seed marketing — Concept, definition and purpose, importance and promotion of quality seed, formal and informal seed supply systems, Seed marketing intelligence and product mix, sales promotion, distribution channels, marketing costs and margins; packaging and labeling, Seed Associations, Factors influencing seed marketing, Seed marketing programs, Seed industry organizations, Marketing of public versus private players, Demand and supply of seed; role of seed replacement rate (SRR), seed multiplication ratio (SMR), economics of seed production; determining seed needs, Seed pricing and price policy, seed processing and /packaging, demand forecasting and factors affecting demand for seeds, effect of price and farm income on seed demand.

Biotechnology in Seed Technology: History of plant tissue culture, Laboratory organization, Composition of nutrient medium, Micro-propagation, Axillary bud proliferation approach, Meristem and shoot tip culture, Bud culture, Advantages of Micro-propagation, Problems associated with micro-propagation, Synthetic seed production, Types of synthetic seeds, methods of development of synthetic seeds, Components of nutrient media for synthetic seed development, Storage of synthetic seeds, Advantages and limitations of synthetic seed production. Planning of seed production, requirements for different classes of seeds in field crops — unit area and rate.

Operation and handling of mechanical drying equipment; effect of drying temperature and duration on seed germination and storability seed processing equipment; seed treating equipment.

Seed production in cross pollinated crops with special reference to land, isolation, Planting ratio of male and female lines, synchronization of parental lines and methods to achieve synchrony; supplementary pollination, pollen storage, hand emasculation and pollination in tomato, Hybrid seed production in Maize, detasseling in maize, identification of rogues and pollen shedders, Pollen collection, storage, viability and stigma receptivity; gametocide application and visits to seed production plots etc., Visit to seed processing plant and commercial controlled and uncontrolled seed stores, Seed industries and local entrepreneurships visit to nearby areas, Different methods of examination of seeds to assess seed-borne microorganisms and to quantify infection percentage, detection of seed borne fungi, bacteria and viruses, identification of storage fungi, control of seed borne diseases, seed treatment methods., Maintenance of aseptic conditions and sterilization techniques, Preparation of nutrient stocks for synthetic media, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Inoculation of explants for micro-propagation, Inoculation of explants for callus induction and subsequently regeneration of plantlets from matured seeds of field and horticultural crops, Synthetic seed preparation.

SKILL ENHANCEMENT COURSES

- A student admitted into 1 st year of B.Sc. Ag. (Hons.) Natural Farming degree program will take 2 skill enhancement courses each of 2 credits in each semester of first year.
- Likewise, the student continuing his/her study into 2nd year of B.Sc. Ag. (Hons.) Natural Farming will undergo 2 skill enhancement course each of 2 credits in each of the 2 semesters of 2nd year.
- The student can select these courses from a basket of skill enhancement courses as indicated below or courses offered by a College.
- The courses maybe offered as module of complementing courses to help the student to achieve skill in a specific area of his interest.
- A college will select such skill enhancement courses in which it has strength as well as there is prospect of local employment and entrepreneurship development.

Indicative List of Skill Enhancement Courses

S. No.	Course Title	Credits
	Natural Farming for Crops, Vegetables and Orchard	2 (0+2)
2.	Production Technology of Bio-inputs	2 (0+2)
3.	Business Management in Natural Farming	2 (0+2)
4.	Seed Production and Certification under Natural Farming	2 (0+2)
5.	Agro-Eco Tourism in Natural Farming	2 (0+2)
6.	Animal Feed Management under Natural Farming	2 (0+2)
7.	Analytical Techniques and Instrumentation for Soil Health Card	2 (0+2)
8.	Landscape Designing and Gardening	2 (0+2)
9.	Self Sustainable Kitchen Gardening	2 (0+2)
10.	Commercial Bee-Keeping under Natural Farming	2 (0+2)
11.	Commercial Sericulture	2 (0+2)
12.	Beneficial Insect Farming	2 (0+2)
13.	Bio-fertlizer and Biopesticide (microbials) production	2 (0+2)
14.	Production Technology Of Biocontrol Agents	2 (0+2)
15.	Mushroom Production Technology	2 (0+2)
16.	Post-harvest Processing Technology	2 (0+2)
	Plantation Crop Production and Processing	2 (0+2)
18.	Poultry Production Technology	2 (0+2)
19.	Piggery Production Technology	2 (0+2)
20.	Commercial Horticulture	2 (0+2)
21.	Floriculture and Landscaping	2 (0+2)
22.	Natural Food Processing	2 (0+2)
23.	Agriculture Waste Management	2 (0+2)
24.	Video Production	2 (0+2)

ONLINE COURSES

- The students will have to take a minimum of 10 credits of online courses (as per UGC guidelines for online courses) as a partial requirement for the B.Sc Ag. (Hons) Natural Farming.
  - The online courses can be from any field such as Basic Sciences, Humanities, Commerce, Business Management, Languages including foreign language, Communication skills, Music, etc. and can be taken from SWAYAM, Diksha, NPTEL, mooKIT, edX, Coursera, or any other portal.
  - The objective is to allow the students to groom their passion or strengthen their knowledge and competency in any field beyond prescribed courses.
  - The courses will be non-gradial as separate certificates would be issued by the Institute/ University offering the courses.
  - However, the University/ institute will keep a record of such courses registered and completed by each student and will indicate the title of the (successfully completed) courses in final transcript issued to the student.
  - The courses can be taken during whole span of degree parogram as per choice of students.
  - The courses will be non-gradial (as separate certificates would be issued by the institutes offering the course).
  - The MOOC courses taken by the student will be separately registered/ approved at the University level.
  - The final transcript will indicate the title of courses taken by the student and the total weeks.

List of Suggestive Online Courses

Course Title	Duration	Credits
Digital Media	16 weeks	
Basics of Photography	12 weeks	
Design, Technology and Innovation	8 weeks	
Visual Communication Design for Digital Media Certificate course in Environmental Sustainabili	4 weeks 8 weeks	
Consumer Protection Legislation	8 weeks	
Communication and Extension for Sustainable Development	15 weeks	
Intellectual Property	12 weeks	
Biopsychology	15 weeks	
NGO'S and Sustainable Development	15 weeks	
Counseling Psychology	12 weeks	
Gender Sensitization: Society Culture and Change	16 weeks	
Basics Of Health Promotion and Education Intervention	8 weeks	
Psychology Of Stress, Health and Well-being	12 weeks	
Diet Management in Health and Disease	12 weeks	
Dairy and Food process and products technology	12 weeks	
Thermal Processing Of Foods	12 weeks	
Nano-technology, Science and Applications	8 weeks	1
Food Science and Processing	12 weeks	
Diet Management in Health and Disease	8 weeks	
Human Nutrition and Biochemistry	12 weeks	1
Personality Development and Communication Skills	8 weeks	

List of Suggestive Online Courses

Course Title	Duration	Credits
Personality Development	8 weeks	
Emotional Intelligence	8 weeks	
Yoga Practices 1	12 weeks	1
Yoga Practices2	12 weeks	1
Ethics: Theories and Applications	12 weeks	
Information Sources and Library Services	6 weeks	1
Qualitative Research Methods and Research Writing	12 weeks	1
Food Science & Processing	12 weeks	1
Food Safety & Quality Control	8 weeks	
*Communication Skills and Computer Operations	24 weeks	2
*Human Resource Management	24 weeks	2
*Business Plan and Finance Management	24 weeks	2
S Physical Education, First Aid and Yoga (to be added)	Compulsory Non-credit Course (CNC)	
$ Start-up (to be developed)		

- SAUS will be free to include more Elective courses with approval from their competent bodies.
- S Compulsory Non-Credit (CNC) course for student.
- The list of Non-Credit courses will be uploaded on College/University website. The student will have choice to select courses of his/her choice up to 20 credits.